UI/UX Design Specification

UTeM Micro-Credential Management System (MCMS)

Document Control

Document Information	
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Prepared by	Aliff, Intern
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Disclaimer:

This document is a working draft prepared by an intern as part of a concept and mock-up for a proposed system. It is not an official document of UTeM or CODL. Names, sample records and figures in screens and documents are fictional. Items marked Assumed or TBD have not been confirmed by the system owner. Unauthorised modification, distribution or misrepresentation is prohibited.

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Revision History

Version	Date	Change
1.0	26 Oct 2026	First issue, written from the approved clickable prototype v2: design principles, visual system, interaction rules, 114 screens, 26 flows and 16 printable documents. Includes the corrections from the planner review of the prototype (Section 9).
1.1	27 Oct 2026	Added Appendix B (every screen with its tabs, wizard steps, pop-ups, warnings, side panels and menus), Appendix C (every page of every printable document and report template) and Appendix D (a storyboard of every step of every flow), all captured from the approved prototype. Fixed in the prototype: the Import CSV and Add buttons of the CODL reference data screen were not shown.
1.2	27 Oct 2026	Specification text aligned with Glossary v0.6, SRS v1.1, PRD v1.1 and SDD v1.1 after the cross-document review of 27 October 2026; screen captures (Appendices B to D) not repeated. FL-19 redefined to follow the demonstration scenario of PRD Section 7 (D-31) because the v1.1 flow mixed three offerings; screen and document counts stated as 97 screens and 17 document entries (114 entries); DESIGN.md assumption tied to D-33 and A-10; design prototype and Release 1 prototype defined; open items UA-06 and UA-07 added. No screen, route or other flow ID changed.
1.3	29 Oct 2026	Recorded how the Release 1 prototype follows this design (Glossary and Decision Log v0.8, D-38): a list of ten rows or fewer carries no pager, a detail screen shows its summary and next action first and keeps long detail in tabs or folds (Section 4.5), and requirement-driven screens the design prototype does not show (a participant's results, feedback) follow the same style.

Glossary

All terms of the Glossary and Decision Log v0.6 apply with exactly the same meaning. The table below adds only the design terms used here.

Term	Definition
Confirmation dialog	A pop-up that states the effect of an action and asks the user to confirm it before anything is saved.
Drawer	A panel that slides in from the right to show a record without leaving the current list.
Flow (FL-XX)	A guided walk-through of a process across screens and roles, with a step bar at the bottom of the prototype.
Hotspot	An outline the prototype draws around every clickable element when the viewer presses H.
Persona	A fictional representative user (Section 2.3). Personas and their data are illustrative.
Route	The address of a screen in the prototype, for example `#/dd/verify/DS-2026-2`. Used as the screen's identifier.
Screen ID	Short identifier of a screen in this document, for example DD-12 (Section 6).
Status tag	A small label with a pale tint and dark text showing the status of a record, always with its word.
Toast	A short message at the top right that confirms a finished action and names the next step.

1. Introduction

1.1 Purpose

This UI/UX Design Specification describes how MCMS looks and behaves for each of its users: the visual system, the interaction rules, every screen, every flow between screens and roles, and every printable document. It turns the requirements of the SRS and the product decisions of the PRD into a design that developers can build without guessing.

The specification was written from the approved clickable prototype v2. The prototype is part of the specification: where a screen is described here, the prototype shows it with sample data, and where this text and the prototype differ, this text prevails and the prototype is corrected. Where this document and the SRS differ on a requirement, the SRS prevails.

1.2 Document Conventions

Convention	Meaning
Screen ID	Role prefix and number, for example DD-12 or ST-04. The full list with routes is in Section 6.4.
Route	Prototype address of a screen, written as `#/dd/verify/:id`. A word after a colon is a record identifier.
FL-XX	A flow of Section 7.
Release 1 / 2	Follows the SRS: **SHALL** is Release 1, **SHOULD** is Release 2. Release 2 controls appear disabled and labelled "Release 2".
Design prototype	The clickable HTML prototype v2 that this specification describes (D-31).
Release 1 prototype	The working application of the SDD on seed data with stub and Manual adapters (D-31). It follows this specification; its acceptance script is PRD Section 7.
FR-XX, BR-XX, NFR-XX	Requirement IDs of the SRS v1.1, as published there.
D-XX, A-XX, OI-XX	Decisions, assumptions and open items of the Glossary and Decision Log v0.6.
Dates and money	Written in this document as "26 Oct 2026"; shown to users as DD/MM/YYYY and RM 0.00 (Section 4.9).

> **Note:** Names, records and figures in the screenshots are sample data of the prototype. They are consistent with each other but are not real people or decisions.

1.3 Intended Audience

Reader	Use of this document	Read first
Supervisor and CODL	Check that every user can do their work and that the design keeps the Guide's stages separate.	Sections 2, 6, 7
Developers and AI development agents	Build screens, components and behaviour exactly as specified.	Sections 3, 4, 5, 6
Testers	Derive UI tests from the interaction rules, flows and screen states.	Sections 4, 7, 9
Content and translation	Write and translate interface text consistently in BM and EN.	Sections 4.9, 8

1.4 Scope

The specification covers the Release 1 web application for all roles: participants (students), lecturers and colleague reviewers, deputy deans, the CODL system administrator and the industry expert. It also covers the public website for visitors and employers, and the documents the system prints or exports. The application is web first and responsive down to a phone width of 360 px (NFR responsiveness); there is no native mobile app (SRS 1.4).

Out of scope: the visual design of UCreds, the national ABC platform and other external systems; the SDD's technical design of components; and Release 2 features beyond showing them as disabled.

1.5 References

1. Glossary and Decision Log for the UTeM MCMS, version 0.6, 27 October 2026 (MCMS_FND_V0.6_Oct_2026).
2. Software Requirements Specification, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_SRS_V1.1_Oct_2026).
3. Product Requirements Document, UTeM MCMS, version 1.0, 26 October 2026 (MCMS_PRD_V1.1_Oct_2026).
4. DESIGN.md, design system "ulearn-carbon" (IBM Carbon with uLearn Navy), held in the project folder.
5. MCMS UI design basis v2 (`UIUX/v2/DESIGN-BASIS.md`) and SRS screen inventory (`UIUX/v2/basis/srs-fr-inventory.md`).
6. MCMS clickable prototype v2 (`UIUX/v2/prototype`) and its sample PDFs (`UIUX/v2/samples`).
7. WCAG 2.1; Garis Panduan Micro-Credential UTeM, Versi 1.0 (the Guide).

2. Design Principles and Users

2.1 Design Principles

Seven principles drive every design decision. They come from the Guide's separation of recognition stages, the PRD design principles (PRD 8.2) and the user needs of the personas.

Principle	What it means on screen	Source
One stage, one record, one look	Approval, offering, result verification, completion, credential, QV, ACR, ABC deposit, transfer and redemption are separate decisions with separate screens. No button combines two stages. Verifying results creates module achievements; confirming completion is a later, separate action that makes the system issue credentials.	D-15, BR-16, G 12–14
Land on what needs me today	Staff open on a task list with age and overdue flags; participants open on their learning and credit.	PRD 8.2
Only the authorised role sees a decision	Decision buttons appear only for the role that may decide. CODL has no academic decision anywhere. The Senate never signs in; the deputy dean records its decisions with the meeting reference.	D-11, D-12, D-28, BR-42
Recognition text is generated	Credit, MQF level and QV claims come only from verified records and templates. A SAMC shows "Credit after QV" until QV is recorded; a lapsed QV removes the claim.	D-17, BR-14, BR-22, BR-23
The rule sits beside the field	Rules are checked as the user types; blocking errors stop submission, warnings let the user continue with the warning on record.	PRD 8.2, ir-forms
History is one click away	Every record has a history from the append-only audit trail. Nothing is deleted; a correction creates a new revision.	D-20, BR-38
Honest about credit	Wherever a participant meets credit, the screen says that deposit is storage, not transfer, and that each transfer is decided case by case.	BR-26, BR-31

2.2 Users and Permissions

The design follows the six kinds of user of decision D-28. Table 1 shows what each role can and cannot do in the interface; buttons for actions a role cannot take are not shown at all, rather than shown disabled.

Table 1: *Roles and their permissions in the interface*

Role (persona)	Signs in	Can	Cannot
Student P8 (Amir Hakimi)	SSO	Browse, register and pay (stub), withdraw, view own records, share credentials, give or decline consent to deposit, apply for transfer, view own stackable plan	See anyone else's records
Lecturer P1 (Dr. Aina Zulkifli)	SSO + MFA	Propose and edit, answer review comments, create offerings, complete the lecturer's readiness items, enter results, send for verification, write quality reviews, gather QV evidence, review colleagues' proposals after declaring conflict of interest	Review or decide own proposal; verify results they entered

Role (persona)	Signs in	Can	Cannot
Deputy dean P2 (Hakim Rahman, FTMK)	SSO + MFA	Assign reviewers, verify category, endorse or return, record Senate decisions, confirm readiness, verify results, confirm completion, correct, replace or revoke on a recorded decision, record QV results, confirm quality reviews, confirm ACR, record deposits, decide transfers into FTMK degrees, redeem or release credit, map pathways and plans, periodic review, suspension	Act outside FTMK; issue credentials by hand (the system issues them)
Senate P3	Never	Decides in its meeting	Use MCMS
CODL S1/P5 (Farid Hashim)	SSO + MFA	Users and scoped roles, reference data, settings with history, bilingual templates, UCreds files, adapter status, test mailbox, audit trail, reports, retention	Any academic decision: approve, verify, issue, ACR, deposit, transfer, module credit flag
Industry expert X2 (Ir. Siti Aminah Yusof)	SSO (limited)	Declare conflict of interest, comment on invited proposals	Rate, recommend or decide
Public X3 (Tan Mei Ling)	Never	Catalogue, pathways, how it works, FAQ, verify a credential	See results, identity numbers, email or matric numbers

A person may hold several roles, each with its own scope. Separation of duties is checked per record: a lecturer who is also a reviewer cannot review their own proposal, and a person who entered a result cannot be its only verifier.

2.3 Personas and Sample Data

The personas of PRD Section 3 appear in the prototype with consistent sample data. The world is set on 26 October 2026 at FTMK, with records at different stages so that every flow can be walked: the UMC Data Structures (BITS 1123, 3 credits, 120 hours) with cohorts from 2025/2 to 2027/1, the SAMC Cloud Native Development (2 credits, MQF level 4, Quality-Verified to 6 December 2026), the MCE Leadership Essentials (16 hours, no credit), and Amir's two deposited or pending credits. Figures add up across screens, for example the 28 learners of Data Structures 2026/2, of whom 27 pass.

3. Visual Design System

MCMS uses the design system of DESIGN.md, "ulearn-carbon": IBM Carbon geometry and type with uLearn Navy as the single brand colour. This section fixes how that system is applied to working screens. Figure 1 shows the result on the deputy dean's task queue.

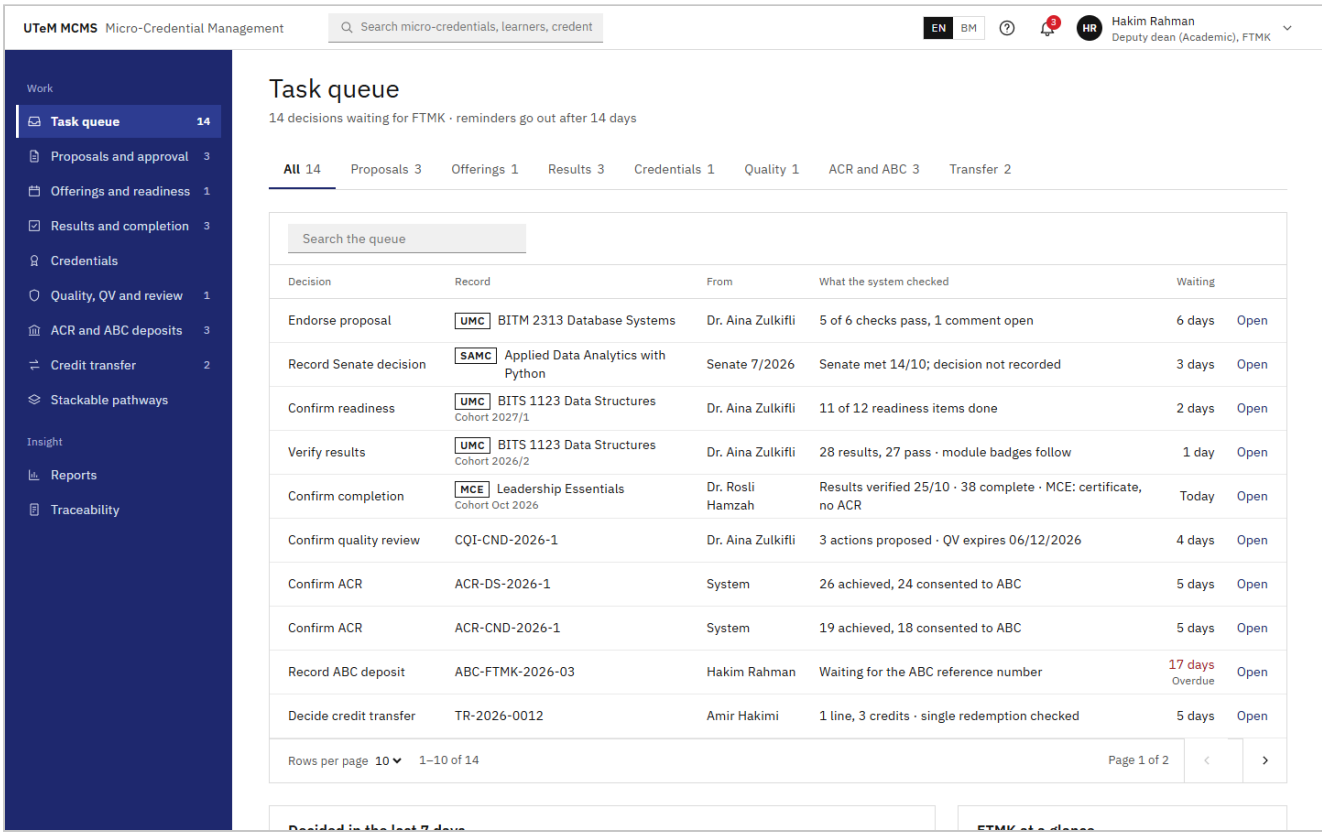


Figure 1: *Application shell: top bar, navy side navigation and white content area (deputy dean task queue, DD-01).*

3.1 Colour

Table 2: *Colour roles*

Role	Value	Use
Primary (uLearn Navy)	#1e2a6e	Side navigation background, the one filled primary button per screen, links, focus outline, selected tab underline
Navy hover / selected	#18225a / #2e3d94	Hover on navigation items; the current page in the side navigation (with a 3 px white left bar)
Ink	#161616	Text, secondary buttons, the dark flow bar
Muted	#525252	Labels, table headers, meta text (meets 4.5:1 on white)
Surface 1	#f4f4f4	Input fields, table bands, hover rows
Hairline	#e0e0e0	Panel borders and table header rule; row rules #ececec
Success / Warning / Error	#24a148 / #f1c21b / #da1e28	Dots, tag tints, notice bars and the danger button only. Never a full background. One red mark per object.

> **Note:** The navy side navigation is the owner's decision of 26 October 2026, overriding the neutral-sidebar guidance of the design craft rules, to give the uLearn brand colour more presence. Content areas stay white.

3.2 Typography

IBM Plex Sans is the only typeface; IBM Plex Mono is used only for identifiers such as setting keys and file names. Every number in a table, KPI or form is set in tabular figures and right-aligned.

Role	Size and weight	Use
Page title	28 / 400	One per screen, sentence case
KPI value	28 / 300	Figures in a KPI strip; light weight per DESIGN.md
Section title	16 / 600	Panel and section headings
Body, cells, navigation	14 / 400, tracking 0.16 px	Everything else
Labels, meta, table headers	12 / 400	Sentence case, grey; never uppercase tracked
Documents	11 pt body, 20 pt light title	A4 printable documents (Section 8)

3.3 Shape, Spacing and Elevation

Corners are square (0 px) for buttons, inputs, panels and modals; only status tags are pill-shaped. Spacing follows a 4 px grid: 24 px between sections, 16 px between fields, 20 px panel padding. Panels are separated by 1 px hairlines and surface change, never by shadows. Shadows are reserved for what floats: menus, toasts, dialogs and drawers.

3.4 Layout and Shell

Region	Specification
Top bar (48 px, white, sticky)	Product name (goes home) · global search (staff) · language switch EN/BM · help · notification bell with unread count · profile button with avatar, name and post
Side navigation (248 px, navy, sticky)	Grouped items with outline icons (16 px) and counts on the right that come from the live task data; the current page is highlighted
Content (white, max 1400 px)	Breadcrumb · title row (title, subtitle, status tag, actions on the right) · content blocks. One filled primary button per screen.
Public site	Top navigation (Home, Catalogue, Pathways to a degree, How it works, FAQ, Verify a credential), Sign in, and a dark footer with contact and policy links
Maintenance banner	A warning bar under the top bar, shown from 48 hours before planned maintenance

3.5 Components

Table 3: *Component catalogue*

Component	Specification
Buttons	40 px high, square. Primary (navy, one per screen), secondary (ink), tertiary (white with navy border), ghost (text), danger (red). A disabled button stays readable and states why, for example "Submit · 1 check failing" or "Confirm ready · 1 item open".
Fields	Label above (12 px), input on Surface 1 with a bottom rule; focus is a 2 px navy outline; error is a 2 px red outline with the message under the field in plain language. Required labels end with an asterisk.
Choice cards	Radio or checkbox inside a bordered card with a title and one-line consequence; the chosen card has a 2 px navy border.
Status tags	Pale tint with dark text of the same hue, always with the status word (Section 4.8). Never above a heading.

Component	Specification
Category mark	UMC, SAMC or MCE in a 1 px ink box beside the micro-credential name.
Tables	12 px grey sentence-case headers, rules between rows, no zebra striping, numbers right-aligned. Search bar above long lists; pager below (Section 4.5).
Tabs	Text tabs with counts; the selected tab has a 2 px navy underline.
KPI strip	One bordered container split by hairlines; label over value; no icons, no sparklines.
Recognition track	A horizontal track of stages (for example Earned, ACR, ABC, Transfer): done stages in ink, the current stage in navy, later stages grey.
Notices	Inline bar with a coloured left edge: info (navy), warning, error, success. Never used as a page background.
Checks list	Dot, statement and one-line evidence; used for system checks, readiness items and rules.
Dialogs, drawers, menus, toasts	See Section 4.

4. Interaction Design

4.1 Every Control Has a Defined Result

Each button, link, tab, table row and menu item does exactly one of the things in Table 4. A control that does nothing is a defect. A control that cannot be used yet is shown disabled with the reason in its label or beside it.

Table 4: *Possible results of a control*

Result	Used for
Go to a screen	Opening a record, the next step of a wizard, a list, a printable document
Confirmation dialog	Every action listed in Section 4.2
Drawer	Reading a record without leaving a list: history, evidence, an email, import problems
In-place change and toast	Saving a draft, marking notifications read, adding a row, accepting evidence
Menu	Notifications, profile, row actions
Download	A real file: signed credential (.json), report and export tables (.csv)

4.2 Confirmation Dialogs

The system asks for confirmation, with a summary of the effect, before: submit, endorse, return, record a decision, verify results, confirm completion (which issues credentials), correct, replace, revoke, confirm ACR, record a deposit, decide a transfer, redeem or release credit, change a setting, assign or remove a role, deactivate an account, import a file, withdraw an enrolment, give or withdraw consent, pay, and sign out with unsaved work (ir-confirm).

Part	Rule
Kicker	The record the dialog is about, for example the credential number
Title	A question naming the action and the count, for example "Verify 28 results?"
Body	What will happen, what will not happen, and who is told. Stage separation is stated where it matters: verifying results does not issue certificates; a deposit is not a transfer; refusing a transfer leaves the ACR and credentials unchanged.
Fields	Any input the decision needs, such as a meeting reference, a reason or an effective date. Required fields are marked and checked.
Footer	Cancel on the left; the action as a verb on the right
Destructive actions	Revoke, deactivate, reject, release, withdraw and terminate use the danger style with a red top edge. The record name appears in the body, a reason is required for the audit trail, and focus starts on a safe control.
Non-blocking warnings	The same dialog states the risk and asks the user to confirm, for example an unresolved non-blocking review comment.

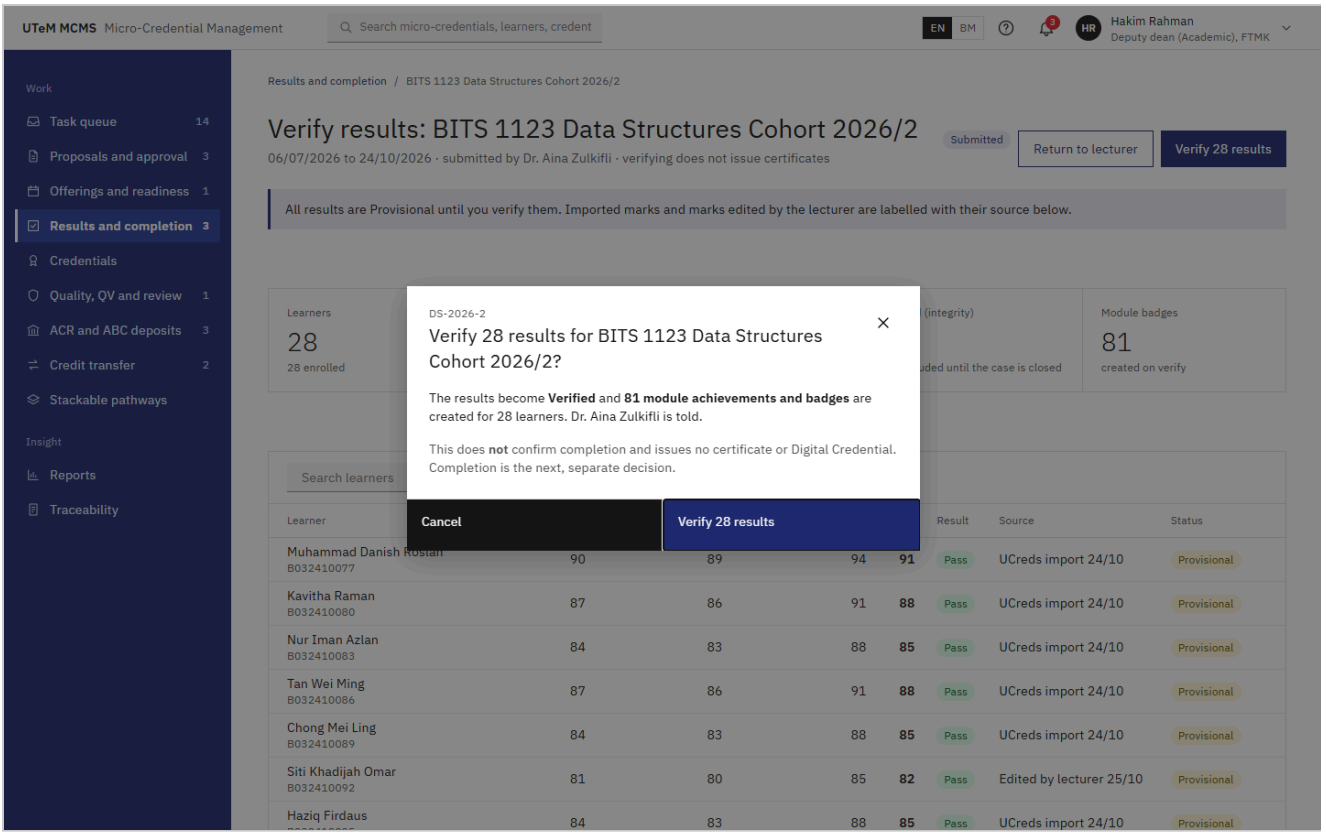


Figure 2: *Confirmation dialog: verifying results states that it creates module achievements only and that completion is confirmed separately.*

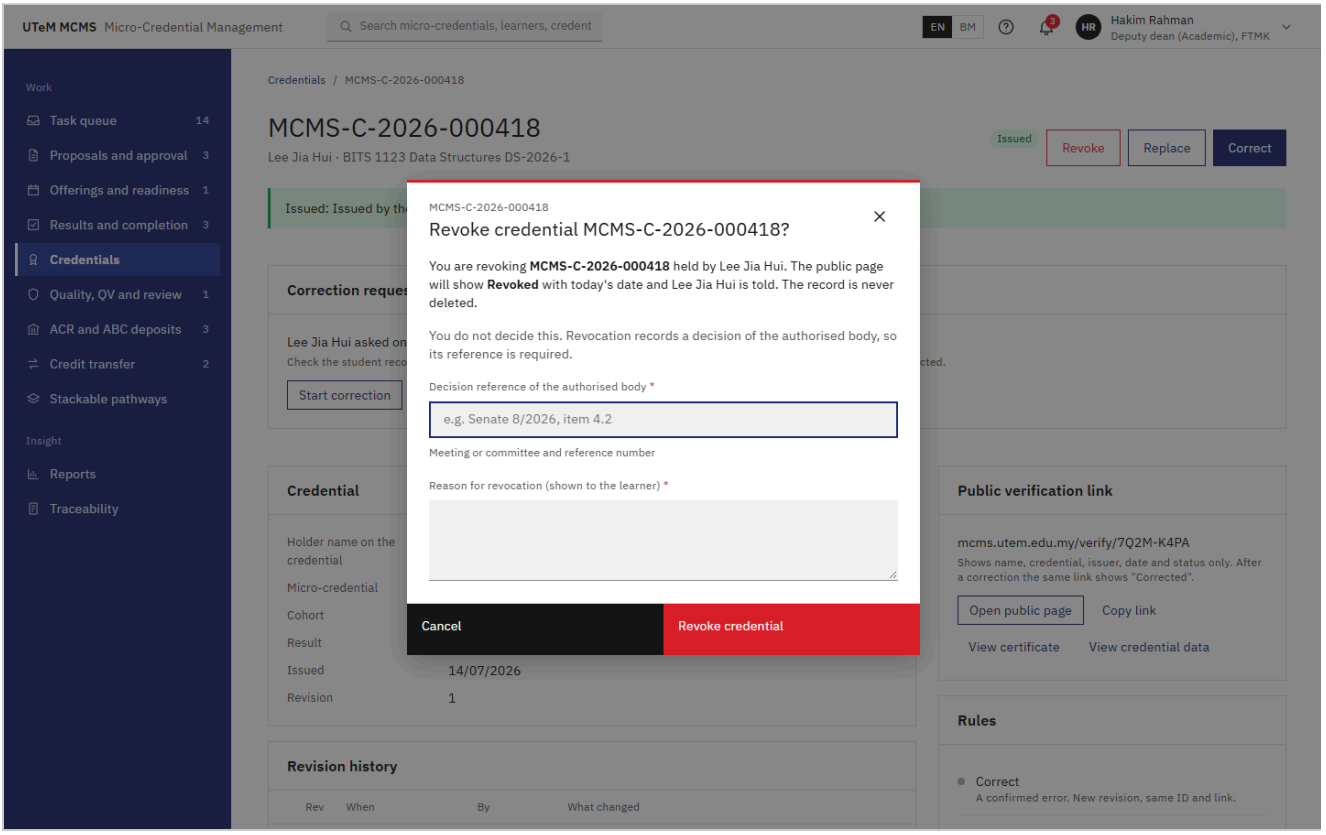


Figure 3: *Destructive dialog: revoking a credential needs the authorised body's decision reference and a reason; the record is never deleted.*

4.3 Feedback

Situation	Design
Action finished	A toast at the top right for 5 seconds: what happened and the next step, for example "Results verified. 27 module badges issued. Confirm completion next."
Field error	The message under the field in plain language; entered data is kept; a toast counts the errors (ir-forms).
System error	Inside the block that failed: what failed, a reference code such as ERR-..., and Retry. No technical detail (nfr-errors).
Long job	Queued, running with a count ("issuing 14 of 27"), done; the user is notified when it finishes (nfr-queue). Used for issuing credentials, imports and large reports.
Empty state	The frame and header stay; one sentence says why it is empty and offers the first action.
Blocked action	The button is disabled and says why; the rule is shown beside the field that causes it.

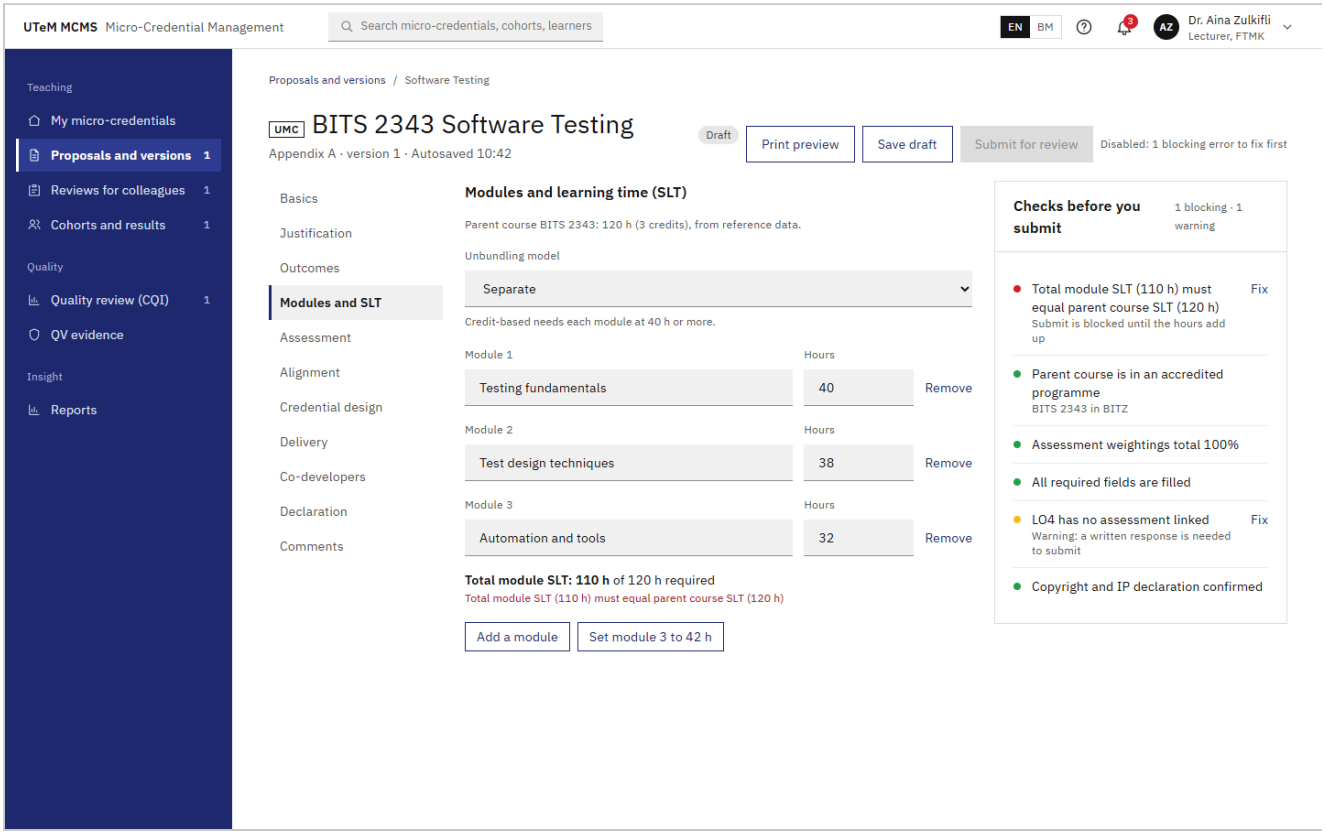


Figure 4: *Rule beside the field: a UMC whose modules total 110 hours cannot be submitted against a 120-hour parent course; the fix is offered in place.*

4.4 Notifications and Profile

The bell in the top bar shows the unread count and opens a panel with the five latest notifications, "Mark all read" and "View all". The notifications page filters by read state, marks a notification read when it is opened and takes the user to the record. Decisions on the user's records, credential actions, consent requests and enrolment changes are mandatory notices that cannot be switched off (fr-ntprefs). Lecturers and deputy deans may choose a daily digest instead of single emails.

The profile button opens a menu: My profile, Notification preferences, Sign-in and sessions, Help and guides, and Sign out (which asks for confirmation). The profile page has four tabs: Details (identity from

SSO, read-only; language, phone and name on certificates editable), Notifications, Sign-in and sessions (MFA, active sessions, idle timeout) and My personal data (view and request a correction, sr-access).

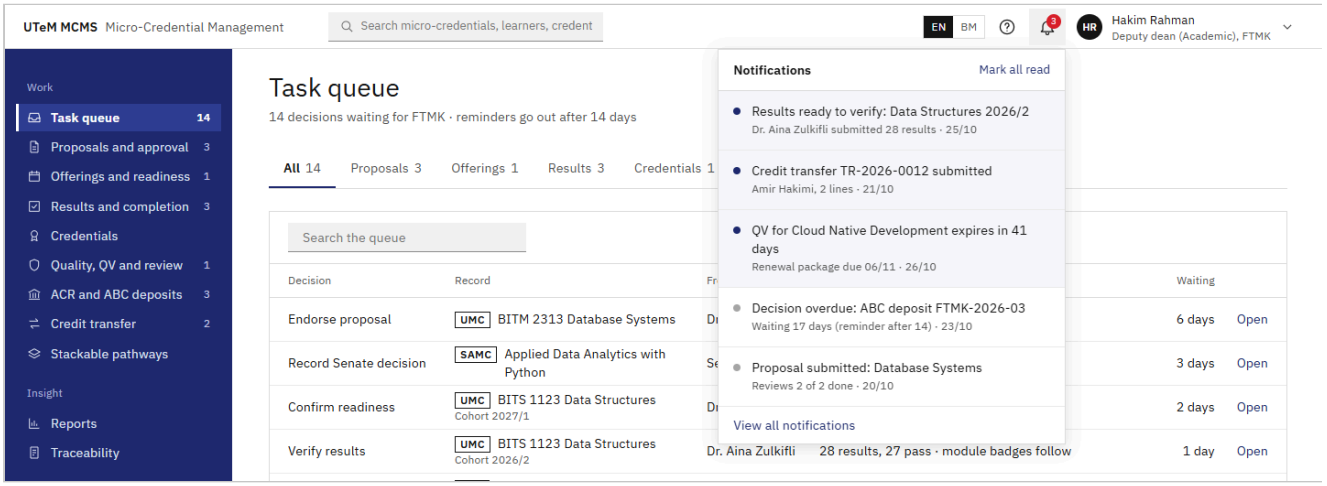


Figure 5: *Notification panel opened from the bell.*

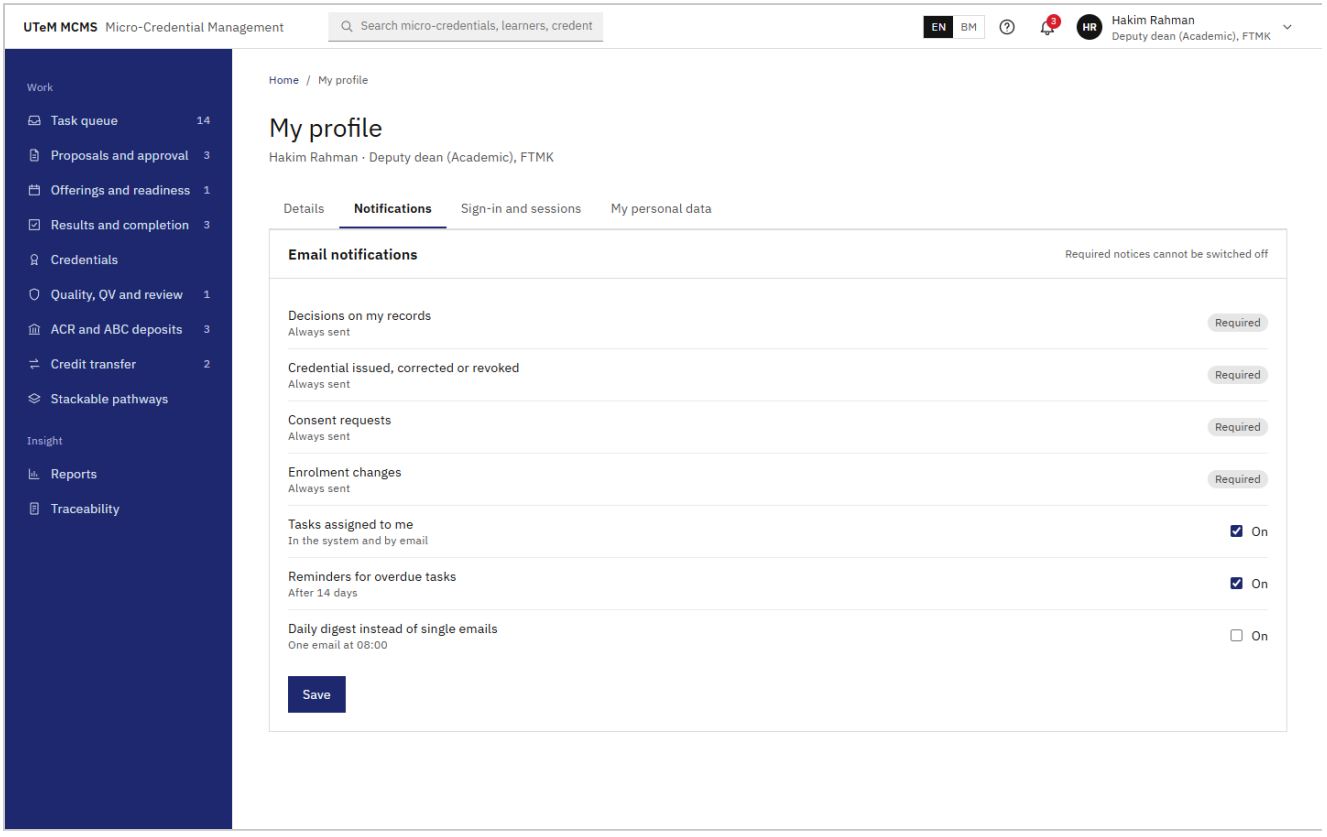


Figure 6: *Profile, notification preferences: required notices are marked and cannot be switched off.*

4.5 Tables and Pagination

Every list that can exceed one page is paginated: rows per page (5, 10, 25 or 50; default 10), a count "1–10 of 28", "Page 1 of 3" and previous and next buttons. Long lists have a search box above the table and filters as selects in the same bar; counts follow the filter. A row that opens a record is clickable across the whole row. A bulk action is one filled button above the table that names the count. Rows that need the user's attention, such as missing marks, are listed first.

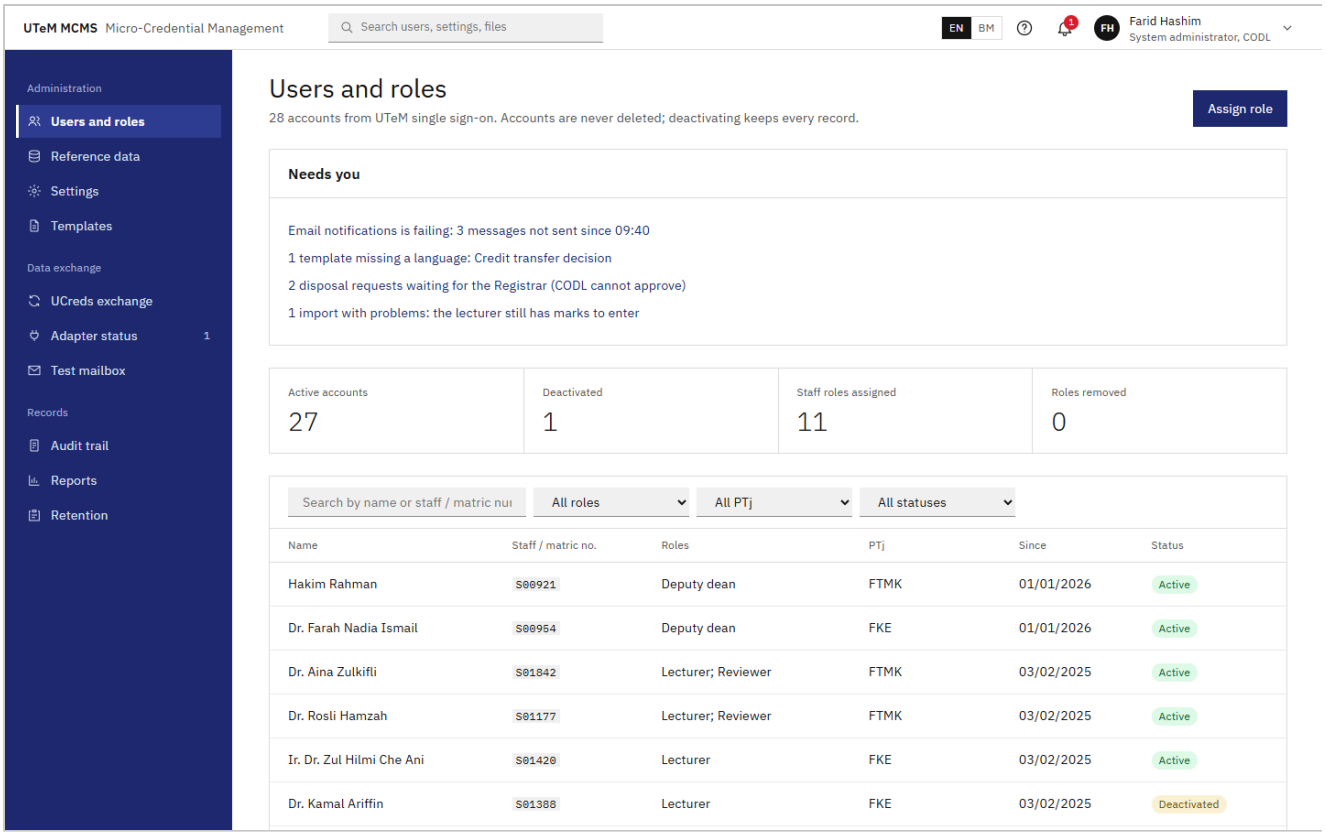


Figure 7: *Table with search, filters and pagination (CODL users and roles), with the "Needs you" panel above it.*

A list of ten rows or fewer shows no search box or pager; its column headers still sort. A detail screen shows its summary and the next action first and keeps long detail in tabs (with counts) or folds that open on request; a tab set opens on the first tab that has something in it. Screens the design prototype does not show but a requirement needs, such as a participant's results, follow the same style (D-38).

4.6 Session, Sign-in and Maintenance

Staff sign in with the UTeM Microsoft account and a second step (MFA); participants sign in without the second step. There is no MCMS password and no public sign-up; an account that is not recognised gets a page explaining how external learners register at Pejabat Pendaftar or HEPA (D-10, A-06). Two minutes before the 30-minute idle timeout a dialog offers "Stay signed in" or "Sign out now". Planned maintenance is announced in a banner 48 hours ahead.

4.7 Accessibility and Language

- Every control works by keyboard: links, tabs, table rows and menu items are focusable, and Enter or Space activates them. Escape closes dialogs, drawers and menus; focus stays inside an open dialog and returns to the control that opened it.
- Every field has a visible label; placeholders never replace labels. Status is never shown by colour alone; every tag carries its word.
- Text meets 4.5:1 contrast; the navy navigation uses white and light-blue text that meets it on #1e2a6e.
- Every screen, notification and credential exists in BM and EN. The language switch changes the language in place and keeps what the user has typed (nfr-langswitch). Bilingual content fields mark a missing language.
- The layout reflows to one column below 760 px; the side navigation becomes a scrollable strip, and tables scroll inside their frame (ir-responsive).

4.8 Status Vocabulary

Statuses use the exact words of PRD 8.3 and the SRS state diagrams. Each status has a short explanation where it first appears on a screen (nfr-plain). Negative statuses are checked before positive ones, so "Not approved" is never shown as an approval.

Table 5: *Status words by record type*

Record	Statuses
Proposal or version	Draft, Submitted, In review, Returned for amendment, Endorsed, Pending approval, Approved, Not approved, Superseded, Suspended, Terminated
Offering	Not ready, Ready, Open, Full, Running, Ended, Cancelled
Enrolment	Registered, Pending payment, Enrolled, Cancelled, Withdrawn, Completed, Not completed
Module or result	Not started, In progress, Provisional, Verified, Held (integrity), Achieved
Completion or credential	Complete, Issued, Corrected, Replaced, Revoked
QV	Not applied, Preparing, Ready, Submitted, Quality-Verified, Expired, Suspended, Withdrawn
ACR or deposit	ACR prepared, Verified, Consent given, Consent declined, Deposited, Correction needed
Credit	Available, Provisionally redeemed, Redeemed, Expired
Transfer	Submitted, Under evaluation, Approved, Partly approved, Not approved, More evidence needed
Stackable plan	Suitable, Needs review, Not yet suitable
CQI action	Open, In progress, Done (effectiveness pending), Closed

4.9 Formats

Dates are shown as DD/MM/YYYY and times as 24-hour Malaysia time (ir-dates, ir-time). Money is shown as "RM 450.00" (nfr-currency). Names are printed in full, including bin, binti, a/l and a/p (nfr-names). Setting keys and file names use the mono face.

5. Navigation

Each role has its own side navigation, grouped by the kind of work, with its home page first. Counts beside an item come from the live task data and fall as work is done. The profile menu, notifications, help and search are the same for every role; search results always stay inside the signed-in role's own screens, so a search never switches role.

Table 6: *Side navigation by role*

Role	Groups and items (home first)
Student	**Learning:** My learning · Find a micro-credential · Registrations and payments. **Achievements:** Achievement record · Credentials. **Credit:** Credit ledger and ABC (count: consents waiting) · Credit transfer · Stackable plan
Lecturer	**Teaching:** My micro-credentials · Proposals and versions · Reviews for colleagues · Cohorts and results. **Quality:** Quality review (CQI) · QV evidence. **Insight:** Reports
Deputy dean	**Work:** Task queue · Proposals and approval · Offerings and readiness · Results and completion · Credentials · Quality, QV and review · ACR and ABC deposits · Credit transfer · Stackable pathways. **Insight:** Reports · Traceability
CODL administrator	**Administration:** Users and roles · Reference data · Settings · Templates. **Data exchange:** UCreds exchange · Adapter status · Test mailbox. **Records:** Audit trail · Reports · Retention
Industry expert	**Invitations:** My review invitations
Public website	Home · Catalogue · Pathways to a degree · How it works · FAQ · Verify a credential · Sign in; footer: contact, Privacy notice (PDPA), Terms of use, Accessibility

Links that lead from one role's screen into another role's work are not offered: for example, CODL is told which marks are missing but cannot open the lecturer's marks screen, and the deputy dean's traceability view does not open CODL's audit screen. Breadcrumbs show the path inside a role.

6. Screens

This section shows the key screens of each role and what they must do. The complete list of 114 screen entries (97 screens and 17 document entries) with their routes is in Section 6.6. Every screen with every tab, wizard step, pop-up, warning and side panel, every printable document page and every flow step is pictured in Appendices B to D of version 1.1, which this version does not repeat. Every screen is in the prototype with sample data; the behaviour of each is described by the SRS requirements listed in `UIUX/v2/basis/srs-fr-inventory.md`.

6.1 Deputy Dean

The deputy dean lands on the task queue (Figure 1): every decision waiting for FTMK, grouped by area, with the system's checks and how long each has waited. Overdue items (14 days or more) are marked in words. Key screens:

Screen	Must do
Proposal review and endorsement	Show system checks, reviews (Support, Amend, Do not support), comments with blocking flags; verify the category as a separate field; assign reviewers when Submitted; Endorse only when no blocking comment is open and the category is verified; Return for amendment with a reason; open the decision pack.
Record Senate decision	Meeting reference and date (not before endorsement), decision (Approved, Approved with conditions, Not approved, Returned for amendment, Deferred), conditions, minutes extract; state that no credit, QV or transfer status is set; for a new version, the transition rule for enrolled learners.
Confirm readiness	Appendix B checklist from the lecturer; accept uploaded evidence; Ready, Ready with conditions or Not ready; the offering opens only after the decision.
Verify results	Learners with provisional results and their source; held results excluded until the case is closed; Verify creates module achievements only.
Confirm completion	Pre-issue checks per learner; failures listed with reasons and not issued; progress of the issuing job; MCE gets a certificate and no ACR; SAMC without QV covering the cohort gets the credential without the QV claim.
Credential	Revision history; Correct (new revision, same link), Replace (new credential, old shows Replaced) and Revoke (decision reference of the authorised body and a reason).
ACR and deposits	Confirm ACR per batch; deposit queue of consented, verified ACR only; export for ABC; record the ABC reference, giving credit valid for 10 years.
Credit transfer	Outcome-by-outcome comparison per line; Approve blocked for a line with no outcome match or no comparison; decisions Approved, Partly approved, Not approved, More evidence needed; redemption and release of credit.
Stackable pathways and plans	Map micro-credentials to courses (MCE refused); individual plans with the 30% UTeM credit and residency checks, stating that 30% does not mean 70% transfers.
Quality, QV and review	QV records with validity and history; QV result (no more than 3 years); quality reviews with action effectiveness; periodic review outcomes; suspension or termination with a transition plan.

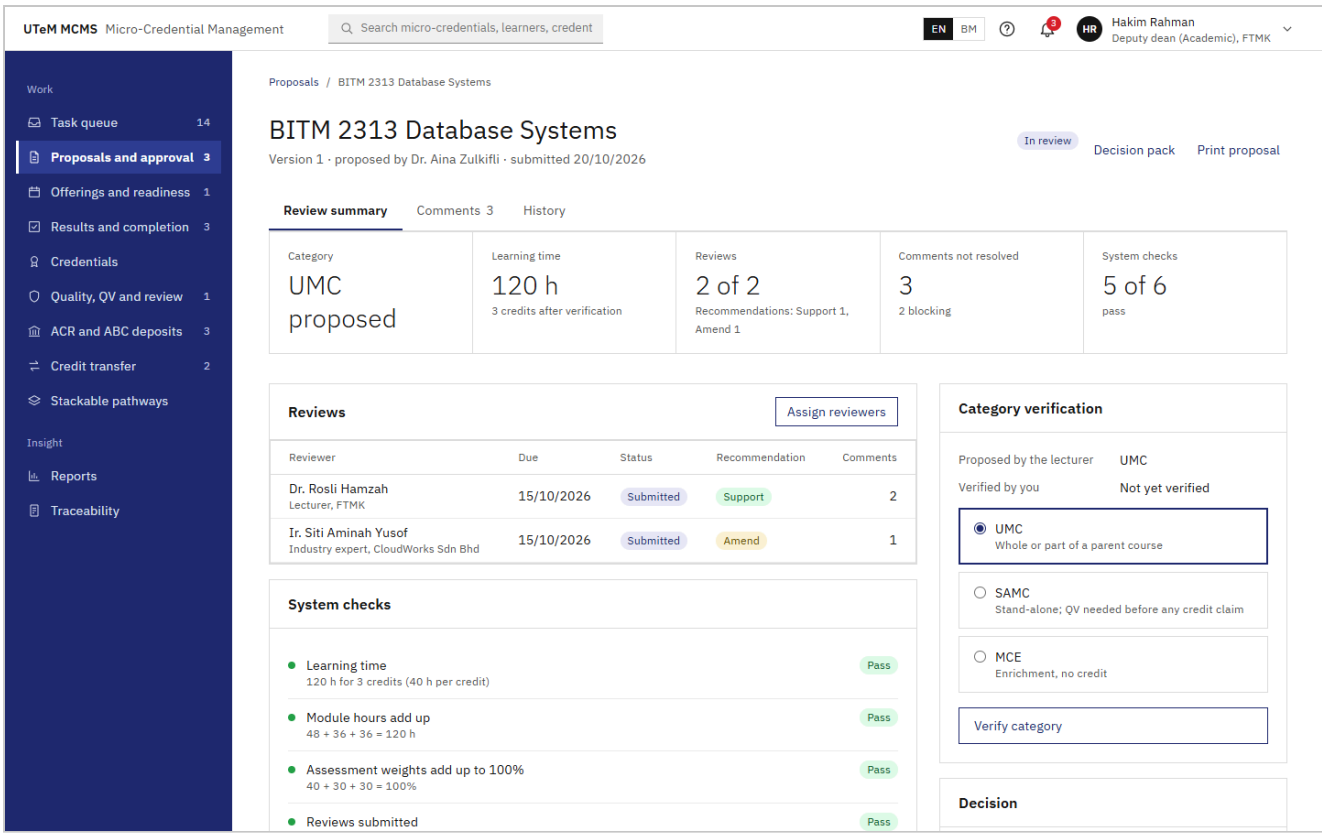


Figure 8: *Proposal review and endorsement (BITM 2313 Database Systems).*

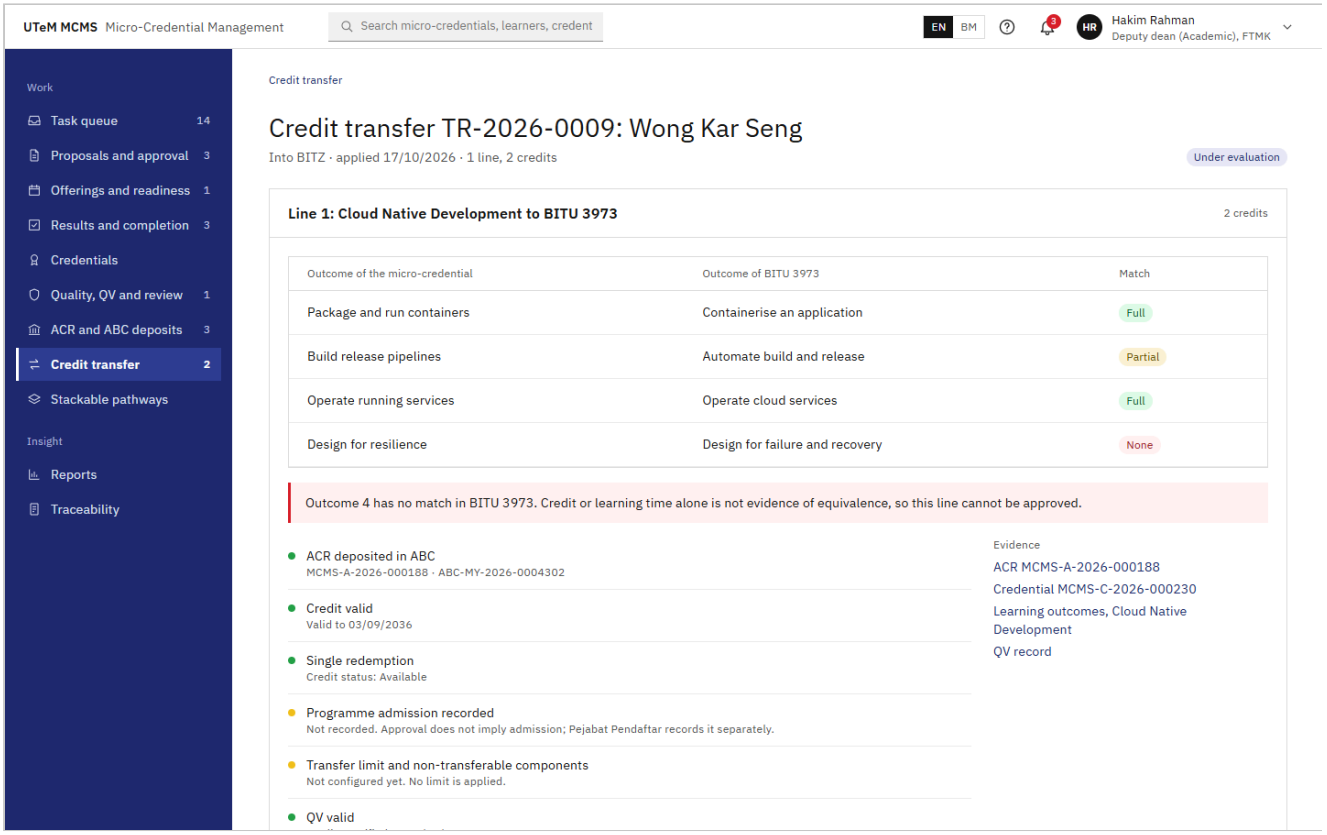


Figure 9: *Credit transfer: a line whose fourth outcome has no match cannot be approved.*

6.2 Lecturer and Reviewer

Screen	Must do
My micro-credentials	"Needs you" list (draft to finish, results to send, quality review due, review invitation, readiness item, QV expiry) and all owned micro-credentials with stage.
New proposal	Three questions decide UMC, SAMC or MCE and explain what the category means for the proposal.
Proposal workspace	Appendix A sections with live checks; SLT must equal the parent course for a UMC; weightings must total 100%; recognition fields follow the category (an MCE has no credit); comments tab to answer reviewers; Submit only when checks pass.
Change to an approved version	Minor or material; a material change becomes a new version submitted for review while learners stay on their version.
Review a colleague's proposal	Conflict of interest first, then the proposal; rate items; recommend Support, Amend or Do not support.
Cohort results	Results from UCreds marked Provisional; missing marks listed first and required; integrity flag; Send for verification; the lecturer cannot verify their own entries.
Readiness, quality review, QV evidence	Lecturer's readiness items with evidence upload; quality review with indicators and actions closed only after an effectiveness check; QV evidence checklist and "Ready to submit" gated by a completed cohort.

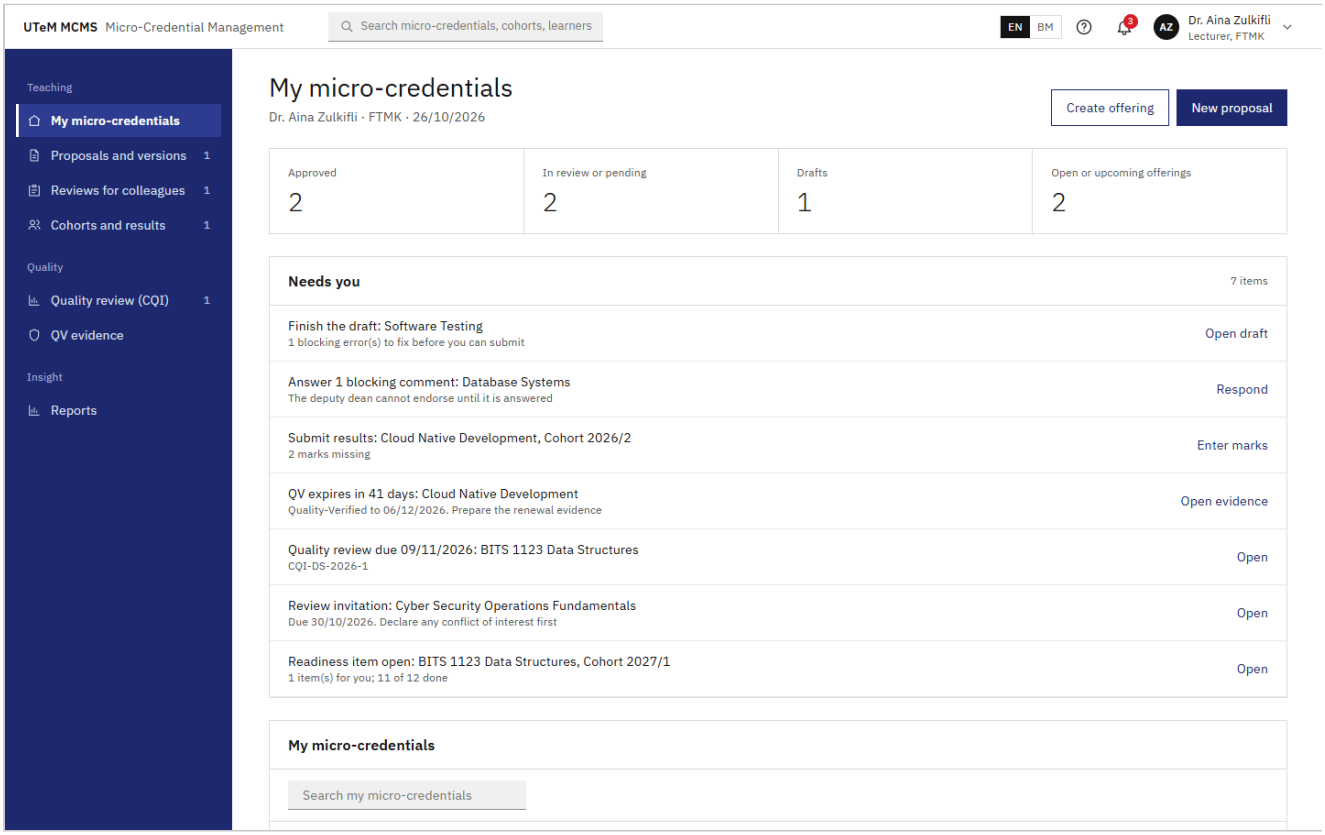


Figure 10: *Lecturer home: what needs Dr. Aina today, then her micro-credentials.*

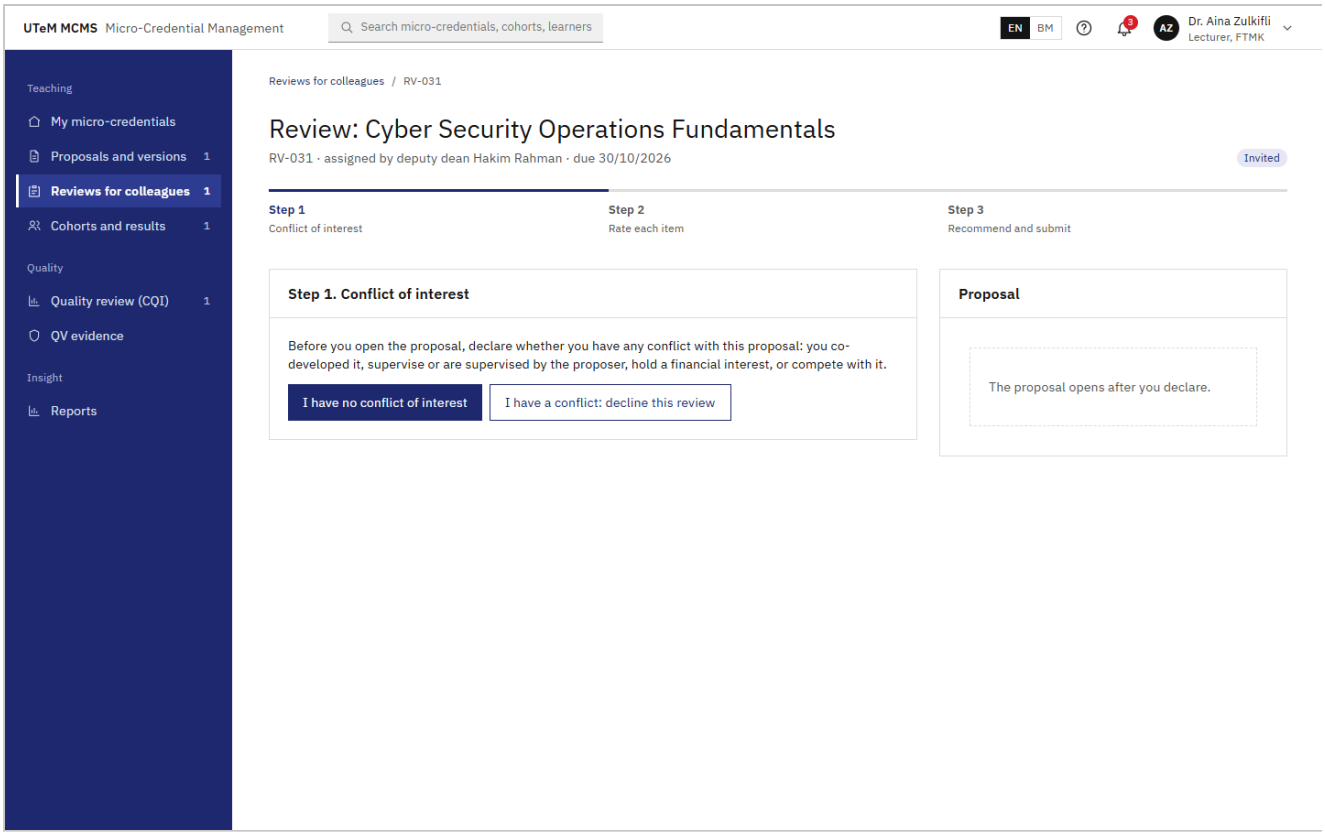


Figure 11: *Colleague review: the proposal opens only after the conflict-of-interest declaration.*

6.3 Student

Student screens are web screens in the same shell as staff, and reflow to a phone.

Screen	Must do
My learning	In progress with module progress, waiting for verification or completion (worded from the cohort's real state), achieved, and actions for the student.
Register	Four steps: eligibility (each rule shown with its result), acknowledgement of the recognition statement and PDPA notice (version and date recorded), dummy payment (failure and retry, never a double charge, capacity checked), confirmation with UCreds access and receipt.
Registrations and payments	Status, receipt, withdraw before the deadline, cancel a pending payment.
Achievement record and credentials	Achievement record labelled "Internal record, not an Academic Credit Record"; credentials with PDF certificate, signed credential, share link and revision history; Replaced and Revoked shown honestly.
Credit ledger and ACR	One ledger with Available, Provisionally redeemed, Redeemed and Expired; consent to deposit with Give, Decline or Postpone; withdrawing consent asks for a reason.
Credit transfer and plan	Apply with deposited, available credit only (MCE and used credit not selectable, with the reason); decision and reason shown; upload more evidence when asked; stackable plan from real records.

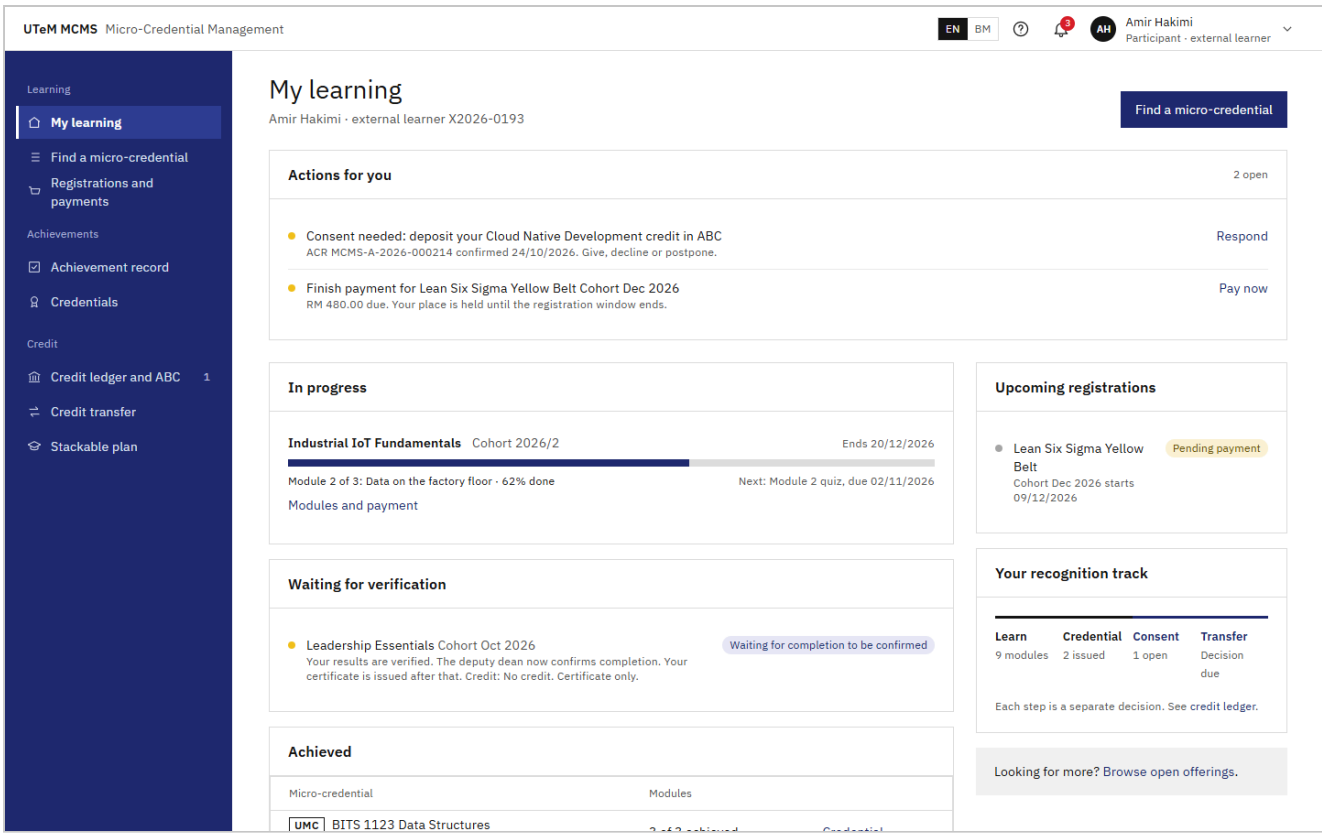


Figure 12: *Student home (Amir Hakimi).*

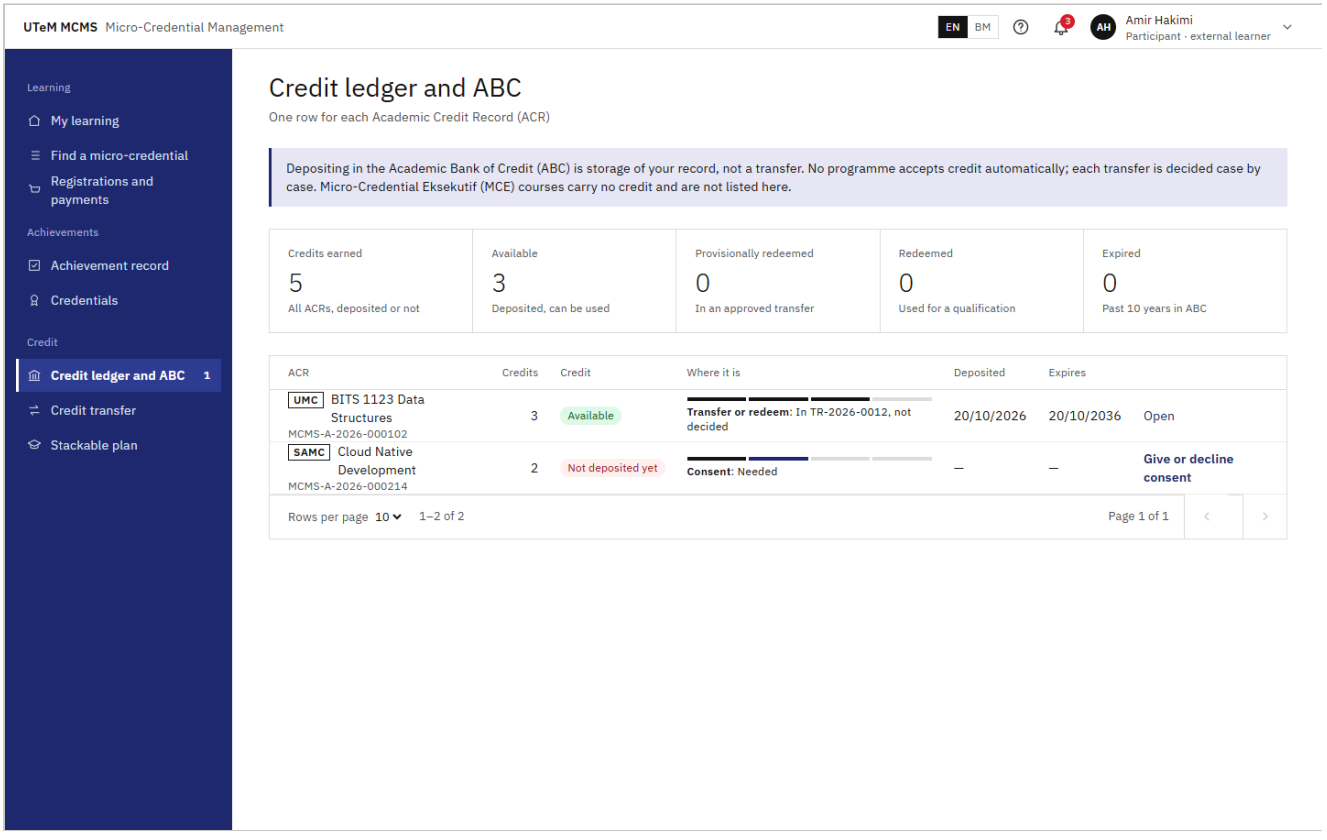


Figure 13: *Credit ledger: one row per ACR with its stage and the honest note that deposit is storage, not transfer.*

6.4 Public Website

The public site informs before it sells. The home page explains the three categories, features the next open offerings, explains how recognition works and links to verification. Offering pages follow the

structure of the UCreds course pages: a facts block (institution, format, effort per week, dates, duration, cost, credit, places) and tabs for Overview, Syllabus, Assessment, Prerequisites, Recognition and FAQ, with an Enrol button. Credit is shown only when it is verified for the cohort's dates; the separate-stages notice states that completion, ACR, deposit and transfer are separate with no guarantee.

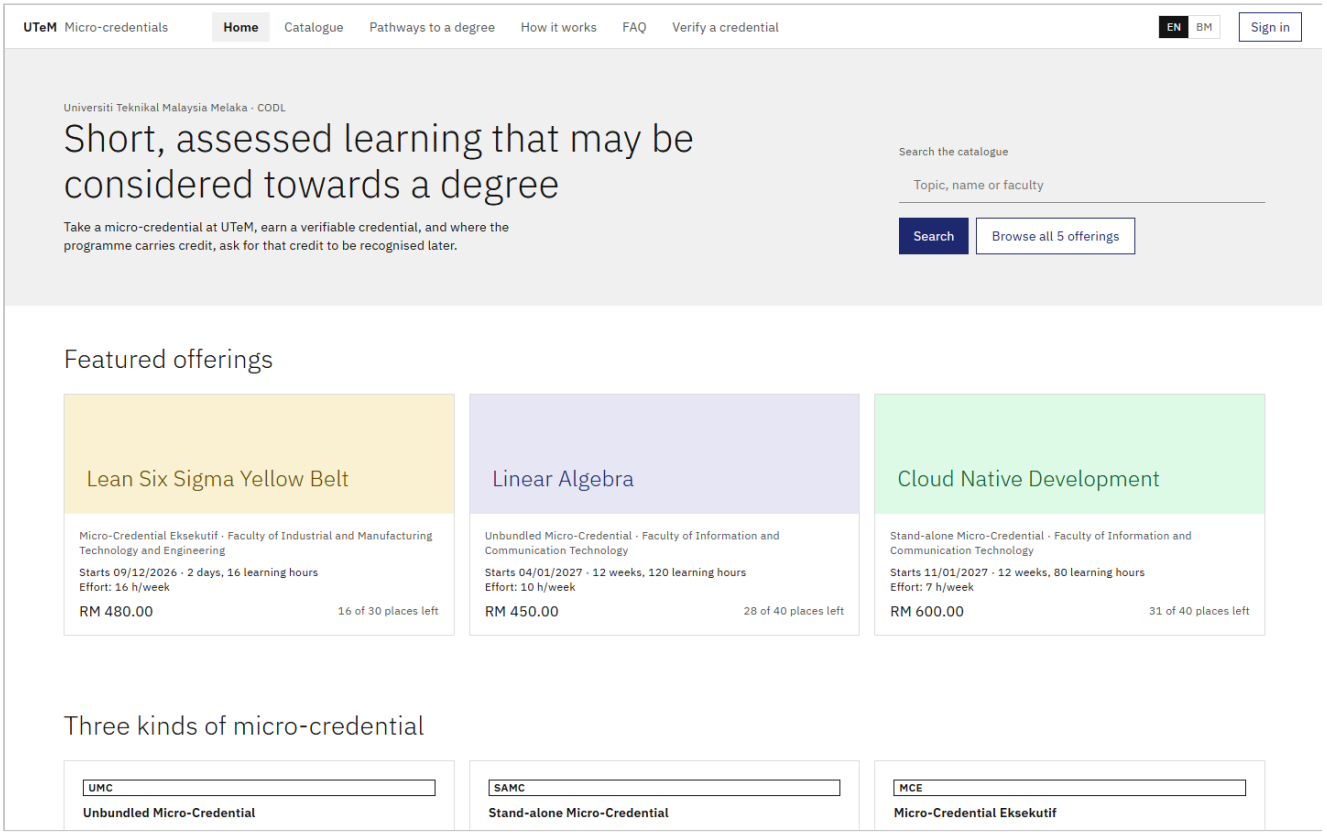


Figure 14: *Public home page.*

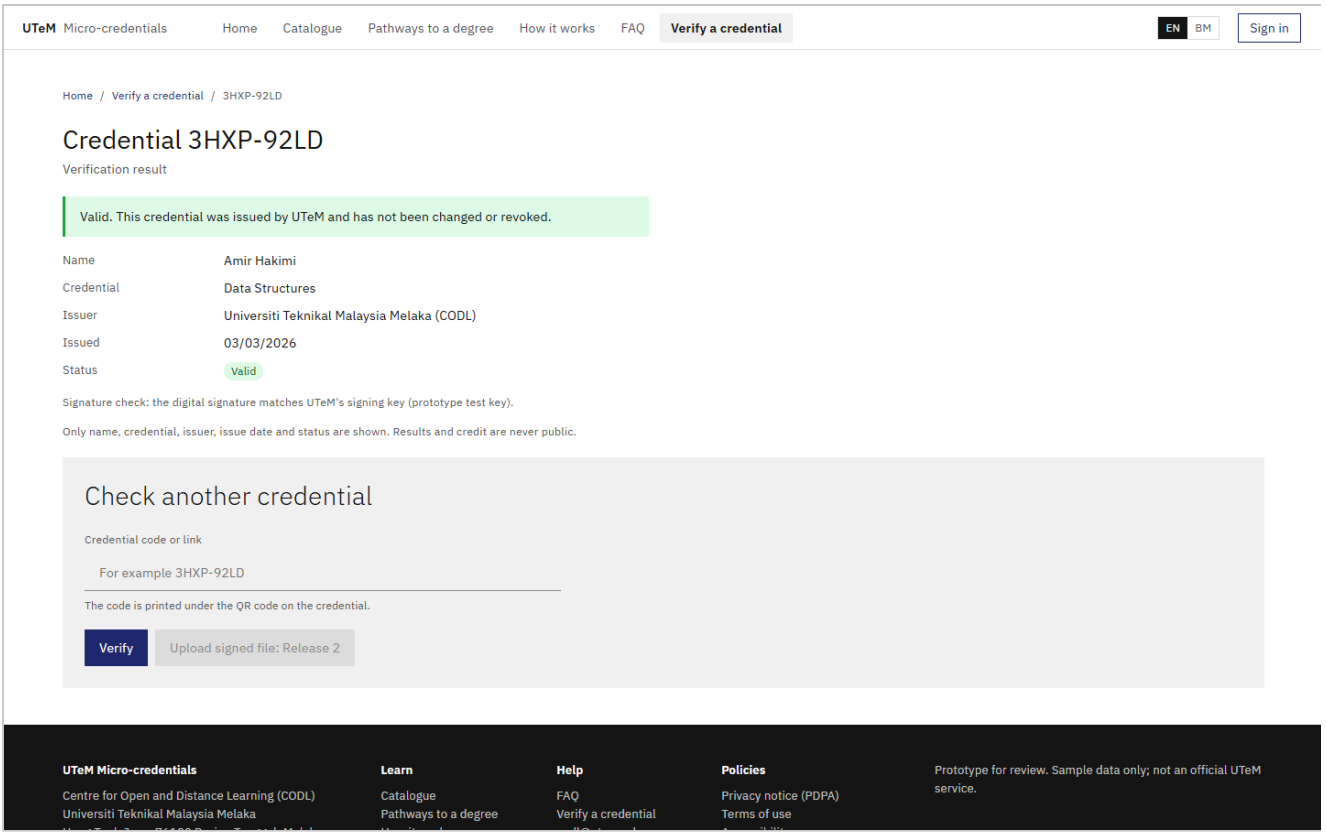


Figure 15: *Public verification: only name, credential, issuer, issue date and status, without signing in. Corrected, Replaced, Revoked and Not found each have their own result.*

6.5 CODL Administrator

CODL lands on users and roles with a "Needs you" panel (failing adapter, templates missing a language, disposals waiting for the Registrar, imports with problems). Role assignment requires a scope, a reason and an effective date; a future date is scheduled; an administrator cannot change their own roles or remove the only deputy dean of a faculty. Settings changes need a reason and a future effective date and keep a history; the public verification fields can only be chosen from the five allowed fields. Templates cannot be published when the BM and EN placeholders differ or when the text makes an unsupported claim such as an automatic transfer. UCreds imports stage rows as Provisional and never create achievements. No CODL screen has an academic decision; where one might be expected, a note names who decides.

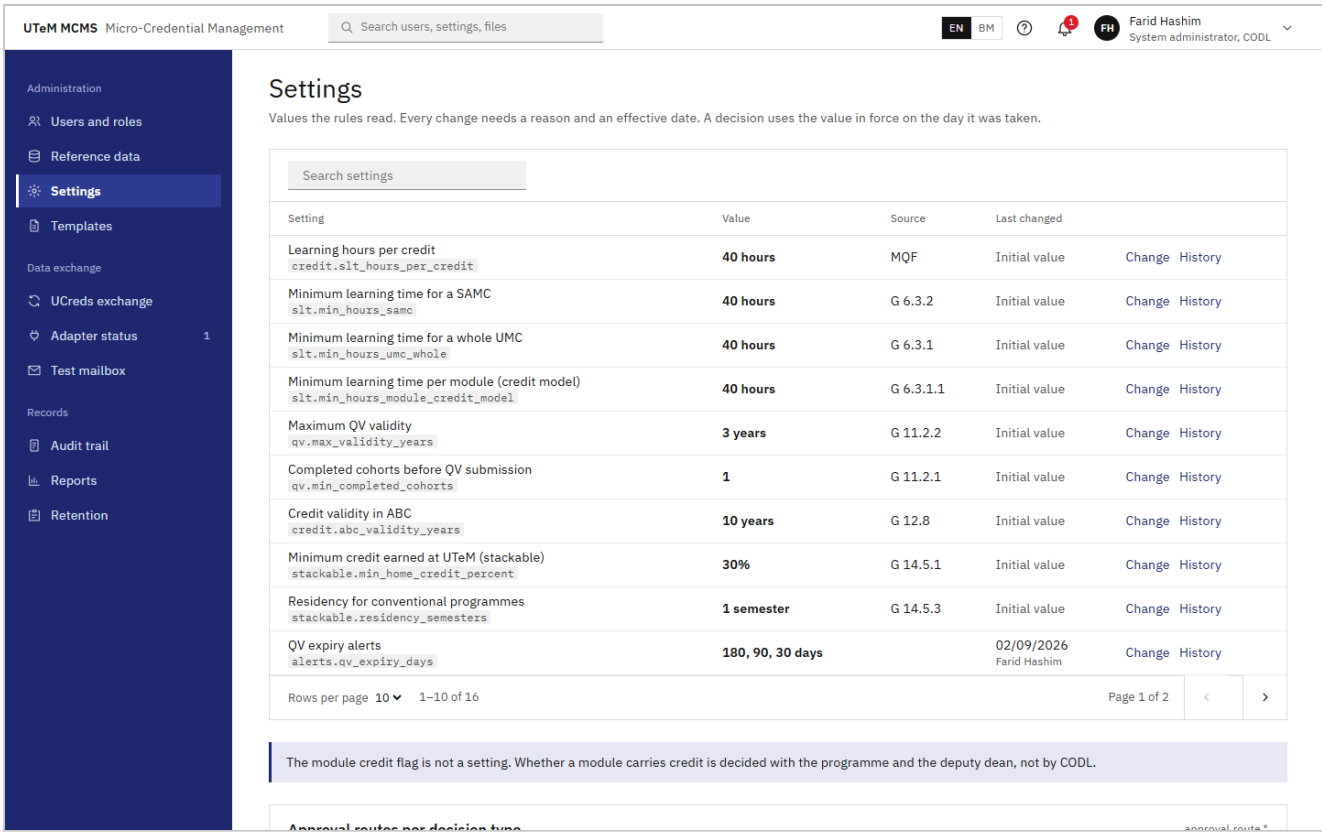


Figure 16: *Settings with source and history; changes take effect from a stated date.*

6.6 Screen Inventory

The 114 screen entries (97 screens and 17 document entries) of the approved prototype, by role. The ID is used in tests and in the SDD; the route opens the screen in the prototype with sample data. Routes with a `:` take a record identifier.

Common and sign-in

Table 7: *Screens for Common and sign-in (10)*

ID	Screen	Route
CMN-01	Personas	`#/`
CMN-02	Sign in	`#/signin`
CMN-03	Verify it is you	`#/signin/mfa`
CMN-04	Account not recognised	`#/signin/noaccount`
CMN-05	Test sign-in	`#/signin/stub`
CMN-06	My profile	`#/profile`
CMN-07	Notifications	`#/notifications`
CMN-08	Help	`#/help`
CMN-09	Search	`#/search`
CMN-10	Something went wrong	`#/error`

Public website

Table 8: *Screens for Public website (11)*

ID	Screen	Route
PUB-01	Micro-credentials at UTeM	`#/public/home`
PUB-02	Catalogue	`#/public/catalogue`
PUB-03	Offering	`#/public/offering/:id`
PUB-04	Pathways to a degree	`#/public/pathways`
PUB-05	How recognition works	`#/public/how`
PUB-06	FAQ	`#/public/faq`
PUB-07	Verify a credential	`#/public/verify`
PUB-08	Verification result	`#/public/verify/:code`
PUB-09	Privacy notice	`#/public/privacy`
PUB-10	Terms of use	`#/public/terms`
PUB-11	Accessibility statement	`#/public/accessibility`

Student (P8)

Table 9: *Screens for Student (P8) (14)*

ID	Screen	Route
ST-01	My learning	`#/st/home`
ST-02	Find a micro-credential	`#/st/catalogue`
ST-03	Register	`#/st/register/:cohort`
ST-04	Registrations and payments	`#/st/enrolments`
ST-05	Registration	`#/st/enrolment/:id`
ST-06	Achievement record	`#/st/record`
ST-07	Credentials	`#/st/credentials`
ST-08	Credential	`#/st/credential/:id`
ST-09	Credit ledger and ABC	`#/st/ledger`
ST-10	Academic Credit Record	`#/st/acr/:id`
ST-11	Credit transfer	`#/st/transfer`
ST-12	New transfer application	`#/st/transfer/new`
ST-13	Transfer application	`#/st/transfer/:id`
ST-14	Stackable plan	`#/st/plan`

Lecturer (P1)

Table 10: *Screens for Lecturer (P1) (18)*

ID	Screen	Route
LEC-01	My micro-credentials	`#/lec/home`
LEC-02	Micro-credential	`#/lec/mc/:id`
LEC-03	Proposals and versions	`#/lec/proposals`
LEC-04	New proposal	`#/lec/new`
LEC-05	Proposal workspace	`#/lec/proposal/:id`
LEC-06	Request a change	`#/lec/change/:id`
LEC-07	Reviews for colleagues	`#/lec/reviews`

ID	Screen	Route
LEC-08	Review a proposal	`#/lec/review/:id`
LEC-09	Cohorts and results	`#/lec/cohorts`
LEC-10	Cohort results	`#/lec/cohort/:id`
LEC-11	Create an offering	`#/lec/offering/new`
LEC-12	Readiness checklist	`#/lec/readiness/:id`
LEC-13	Quality review (CQI)	`#/lec/cqi`
LEC-14	Quality review record	`#/lec/cqi/:id`
LEC-15	QV evidence	`#/lec/qv`
LEC-16	QV evidence	`#/lec/qv/:id`
LEC-17	Reports	`#/lec/reports`
LEC-18	Report	`#/lec/report/:id`

Industry expert (X2)

Table 11: *Screens for Industry expert (X2) (2)*

ID	Screen	Route
EXT-01	My review invitations	`#/ext/home`
EXT-02	Comment on a proposal	`#/ext/review/:id`

Deputy dean (P2)

Table 12: *Screens for Deputy dean (P2) (29)*

ID	Screen	Route
DD-01	Task queue	`#/dd/queue`
DD-02	Proposals and approval	`#/dd/proposals`
DD-03	Proposal review	`#/dd/proposal/:id`
DD-04	Record Senate decision	`#/dd/senate/:id`
DD-05	Decision pack	`#/dd/pack`
DD-06	Offerings and readiness	`#/dd/offerings`
DD-07	Confirm readiness	`#/dd/readiness/:id`
DD-08	Results and completion	`#/dd/results`
DD-09	Verify results	`#/dd/verify/:id`
DD-10	Confirm completion	`#/dd/completion/:id`
DD-11	Credentials	`#/dd/creds`
DD-12	Credential	`#/dd/cred/:id`
DD-13	Quality, QV and review	`#/dd/quality`
DD-14	QV record	`#/dd/qv/:id`
DD-15	Quality review	`#/dd/cqi/:id`
DD-16	Periodic review	`#/dd/review/:id`
DD-17	Suspension or termination	`#/dd/suspend/:id`
DD-18	ACR and ABC deposits	`#/dd/acr`
DD-19	Confirm ACR	`#/dd/acr/:id`

ID	Screen	Route
DD-20	Record ABC deposit	`#/dd/deposit/:id`
DD-21	Credit transfer	`#/dd/transfer`
DD-22	Transfer decision	`#/dd/transfer/:id`
DD-23	Stackable pathways	`#/dd/stack`
DD-24	Pathway mapping	`#/dd/pathway/:id`
DD-25	Individual stackable plan	`#/dd/plan/:id`
DD-26	Reports	`#/dd/reports`
DD-27	Report	`#/dd/report/:id`
DD-28	Traceability	`#/dd/trace`
DD-29	Trace a participant	`#/dd/trace/:who`

CODL system administrator (S1/P5)

Table 13: *Screens for CODL system administrator (S1/P5) (13)*

ID	Screen	Route
CODL-01	Users and roles	`#/codl/users`
CODL-02	User	`#/codl/user/:id`
CODL-03	Reference data	`#/codl/refdata`
CODL-04	Settings	`#/codl/settings`
CODL-05	Templates	`#/codl/templates`
CODL-06	Template	`#/codl/template/:id`
CODL-07	UCreds exchange	`#/codl/ucreds`
CODL-08	Adapter status	`#/codl/adapters`
CODL-09	Test mailbox	`#/codl/mailbox`
CODL-10	Audit trail	`#/codl/audit`
CODL-11	Reports	`#/codl/reports`
CODL-12	Report	`#/codl/report/:id`
CODL-13	Retention	`#/codl/retention`

Printable documents

Table 14: *Screens for Printable documents (17)*

ID	Screen	Route
DOC-01	Payment receipt	`#/doc/receipt/:id`
DOC-02	Enrolment confirmation	`#/doc/enrolment/:id`
DOC-03	Certificate	`#/doc/certificate/:id`
DOC-04	Digital credential	`#/doc/credential/:id`
DOC-05	Achievement record	`#/doc/record/:who`
DOC-06	Academic Credit Record	`#/doc/acr/:id`
DOC-07	Proposal print	`#/doc/proposal/:id`
DOC-08	Senate decision pack	`#/doc/pack/:id`
DOC-09	Report	`#/doc/report/:id`

ID	Screen	Route
DOC-10	ABC deposit export	`#/doc/abc/:id`
DOC-11	QV submission package	`#/doc/qvpack/:id`
DOC-12	Credit transfer decision letter	`#/doc/transfer/:id`
DOC-13	Suspension notice	`#/doc/suspend/:id`
DOC-14	Import validation report	`#/doc/import/:file`
DOC-15	UCreds enrolment export	`#/doc/export/:cohortId`
DOC-16	Audit trail export	`#/doc/audit`
DOC-17	Documents and PDFs	`#/doc/index`

7. User Flows

A flow is a process walked across screens and, often, across roles. Each flow of Table 16 is in the prototype's Flows page; starting one shows a step bar at the bottom of the screen that says what is happening and what to click next. Negative paths have flows of their own, for example a payment failure, a transfer line that cannot be approved, or a credential that was revoked.

7.1 End-to-End Journey

Figure 17 shows the journey of SRS use case A.5 and the demonstration scenario of PRD Section 7 (flow FL-19), in four acts. Amir follows Data Structures (UMC) from proposal to credit used; Cloud Native Development (SAMC) shows Quality Verification and its expiry; Leadership Essentials (MCE) completes with no ACR. FL-19 is the acceptance script of the Release 1 prototype (D-31). Each box is a separate decision by a separate role; the system issues credentials and generates the ACR only after the deputy dean's decision.

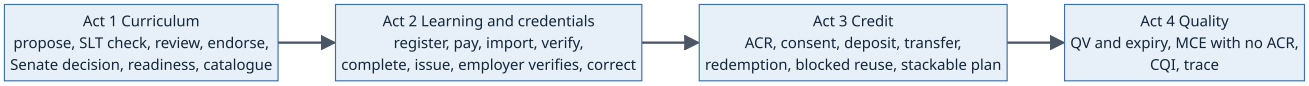


Figure 17: *End-to-end journey (FL-19): four acts, each stage its own decision by its own role.*

7.2 Hand-offs Between Roles

When one role finishes a step, the next role sees it at once: a task appears in the deputy dean's queue, a notification reaches the person affected, and the record's status and history change. Table 15 lists the hand-offs the design guarantees; each is tested end to end in the prototype.

Table 15: *Hand-offs between roles*

From	Action	Lands with	Result seen
Lecturer	Sends cohort results for verification	Deputy dean	Task "Verify results"; every learner's provisional result on the verify screen
Lecturer	Sends the readiness checklist	Deputy dean	Task "Confirm readiness"; uploaded evidence ready to accept; after Ready the offering appears in the public catalogue
Lecturer	Answers a review comment	Deputy dean	Comment shows as Answered and can be resolved; endorsement unblocks when no blocking comment is open
Deputy dean	Confirms completion	Student	Credentials issued; enrolment Completed; notification
Student	Gives consent to deposit	Deputy dean	Task "Record ABC deposit"; ACR in the deposit queue
Deputy dean	Records the deposit	Student	Ledger shows the credit Available for 10 years
Student	Applies for transfer	Deputy dean	Task "Decide credit transfer" with the evidence attached
Deputy dean	Approves a transfer line	Student	Credit Provisionally redeemed; it can no longer be selected for another application
Deputy dean	Asks for more evidence	Student	Upload form on the application; the upload reopens the deputy dean's task
CODL	Imports UCreds results with problems	Lecturer	Notification naming the missing marks; rows needing marks listed first

7.3 Flow Catalogue

Table 16: *Walk-through flows in the prototype*

ID	Flow	Actors	Steps
FL-00	**Tour the deputy dean task queue**. How the queue, tabs, search and pagination behave	Deputy dean	2
FL-01	**Sign in with SSO and MFA, land on the task page**. UTeM Microsoft sign-in, second step, then the task queue	Any staff	3
FL-02	**Propose a UMC, fix the SLT, submit**. Category wizard, a blocked draft at 110 h, the fix to 120 h, and submit	Lecturer	5
FL-03	**Review a colleague's proposal**. Conflict of interest first, then rate, then recommend	Lecturer as reviewer	4
FL-04	**Endorse a proposal, then record the Senate decision**. Endorse is blocked until the category is verified and the blocking comment is resolved	Deputy dean	7
FL-05	**Create an offering and confirm readiness**. The cohort opens only after every Appendix B item is done and you confirm	Lecturer, deputy dean	4
FL-06	**Find, register and pay**. Public offering page to a paid registration for Cloud Native Development 2027/1	Student	6
FL-06b	**Payment failure and retry**. A declined card changes nothing; a second try works and charges once	Student	3
FL-07	**Import UCreds results with a bad row**. Import results-CND-2026-2.csv with 2 problems; the lecturer enters the missing marks and submits for verification	CODL, lecturer, deputy dean	3
FL-08	**Verify results, then confirm completion**. Two separate decisions: verifying gives module badges, confirming completion makes the system issue credentials	Deputy dean	3
FL-09	**Share a credential and have it verified**. Share link, a genuine check, a revoked credential and a code that does not exist	Student, public	4
FL-10	**Correct a credential name**. A correction makes a new revision; the original is kept and the public link shows Corrected	Deputy dean	2
FL-11	**Consent to ABC deposit, confirm ACR, record deposit, see ledger**. Amir gives consent, Hakim finds the ACR in the deposit queue and records the deposit, and the ledger shows the credit	Student, deputy dean	4
FL-12	**Apply for credit transfer; decide by outcome; provisionally redeemed; second use blocked**. Amir applies, Hakim approves, the credit is held, and a second case with an unmatched outcome is blocked	Student, deputy dean	5
FL-13	**Stackable plan: 25% "Not yet suitable" and a suitable plan**. Two individual plans for BITZ, one below 30% UTeM credit and one that is suitable	Deputy dean	4
FL-14	**QV evidence, gate, record QV result; expiry removes claims**. Evidence on the lecturer side, then the deputy dean builds the package, records the submission and the result	Lecturer, deputy dean	4
FL-15	**Quality review: close an action only after the effectiveness check**. The lecturer view, then submit to the deputy dean	Lecturer, deputy dean	4
FL-16	**Suspend an offering with a transition plan**. Periodic review leads to a suspension with a transition plan; earned credentials stay valid	Deputy dean	4

ID	Flow	Actors	Steps
FL-17	**Assign a scoped role and change a setting**. Give a role with scope, reason and date; change a setting with a reason; see its history and the audit trail	CODL administrator	4
FL-18	**Trace a participant from approval to credit use; audit trail**. Follow Amir through every stage, then read the audit rows for the same records	Deputy dean	3
FL-19	**End-to-end: the four acts of the demonstration scenario**. SRS use case A.5 and PRD Section 7: Data Structures from proposal to credit used, Cloud Native Development Quality Verification, Leadership Essentials with no ACR; every recognition stage as its own decision	All roles	4 acts
FL-20	**Industry expert comments on a proposal**. Conflict of interest, then comment only	Industry expert	3
FL-21	**Withdraw from an enrolment**. Withdraw with a reason; achieved modules stay on record	Student	3
FL-22	**A visitor explores the public site**. From the home page to a filtered catalogue, an offering, how recognition works and pathways	Public visitor	5
FL-23	**An employer verifies a credential**. Valid, corrected, revoked and not-found results	Tan Mei Ling, HR manager (public)	7
FL-24	**Documents and PDFs**. Open the printable documents: receipt, certificate, Digital Credential, full report, ABC export.	All roles	8

8. Printable Documents and Content

8.1 Printable Documents

Every download, print or export in MCMS opens a document laid out on A4 pages, which the user can print or save as PDF. Where a machine-readable file is needed, the document also offers a real download. Each page has a header with the UTeM name, the document title, reference and date, and a footer with the reference, page number, generation time and "Generated by MCMS". Documents for participants are bilingual, with BM beneath EN (D-02). No identity number, email or marks appear on documents that leave UTeM.

Table 17: *Printable documents*

Document	Opened from	Content	File
Payment receipt	Student registrations	Receipt number, payer, offering, fee breakdown, method, reference, total; "No payment received" when unpaid	PDF
Enrolment confirmation	Student registrations	Access to UCreds, dates, withdrawal deadline, recognition statement, separate-stages notice	PDF
Certificate	Student and deputy dean credential pages	Landscape, bilingual; holder, micro-credential, category, verified credit only, credential ID, verification code and link, QR, two signatures; Revoked band or revision line when relevant	PDF
Digital Credential	Credential pages	Signed credential shaped on Open Badges 3.0 / W3C VC with a summary and proof	JSON, PDF
Achievement record	Student, profile	Labelled "Internal record, not an Academic Credit Record"; enrolments, modules, badges, completions	PDF
Academic Credit Record	Ledger, deposit screens	Credits, MQF level, completion, verifier, consent, ABC reference and expiry; storage, not transfer	PDF
Proposal (Appendix A)	Lecturer workspace, deputy dean review	All Appendix A sections with SLT totals, weightings, alignment, version and status	PDF
Senate decision pack	Deputy dean proposal review	Cover, summary, category verification, reviews, comment resolution, endorsement, space for the minute reference	PDF
Report (10 kinds)	Reports	Cover, scope, period, filters, summary figures, tables that add up, method notes, page numbers	PDF, CSV
ABC deposit export	ACR and deposits	Batch header, one row per ACR, totals, format note	PDF, CSV
QV submission package	QV records	Cover, evidence index, cohort results, quality reviews, declaration	PDF
Transfer decision letter	Transfer case	Outcome per line, credits approved, conditions, statement that refusal leaves ACR and credentials unchanged	PDF
Suspension notice	Suspension	Notice to participants with the transition plan; earned credentials stay valid	PDF
UCreds import report	UCreds exchange	File, row counts, rows staged as Provisional, rejected rows with reasons	PDF, CSV
Enrolment export	UCreds exchange, lecturer cohort	Template version, columns, one row per enrolled learner	PDF, CSV
Audit export	Audit trail	Entries with an integrity line	PDF, CSV

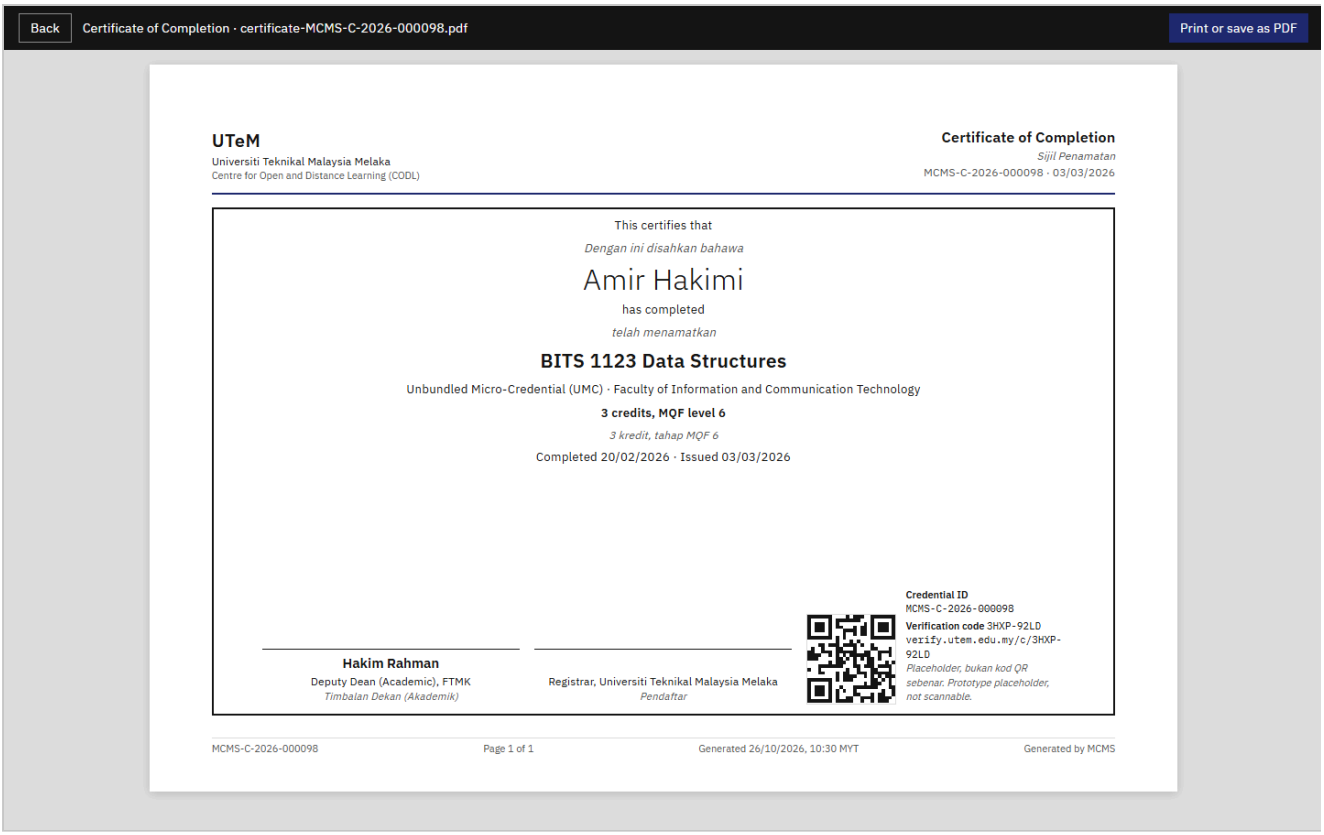


Figure 18: *Certificate document with the print bar (sample PDFs of every document are in* `UIUX/v2/samples`*).*

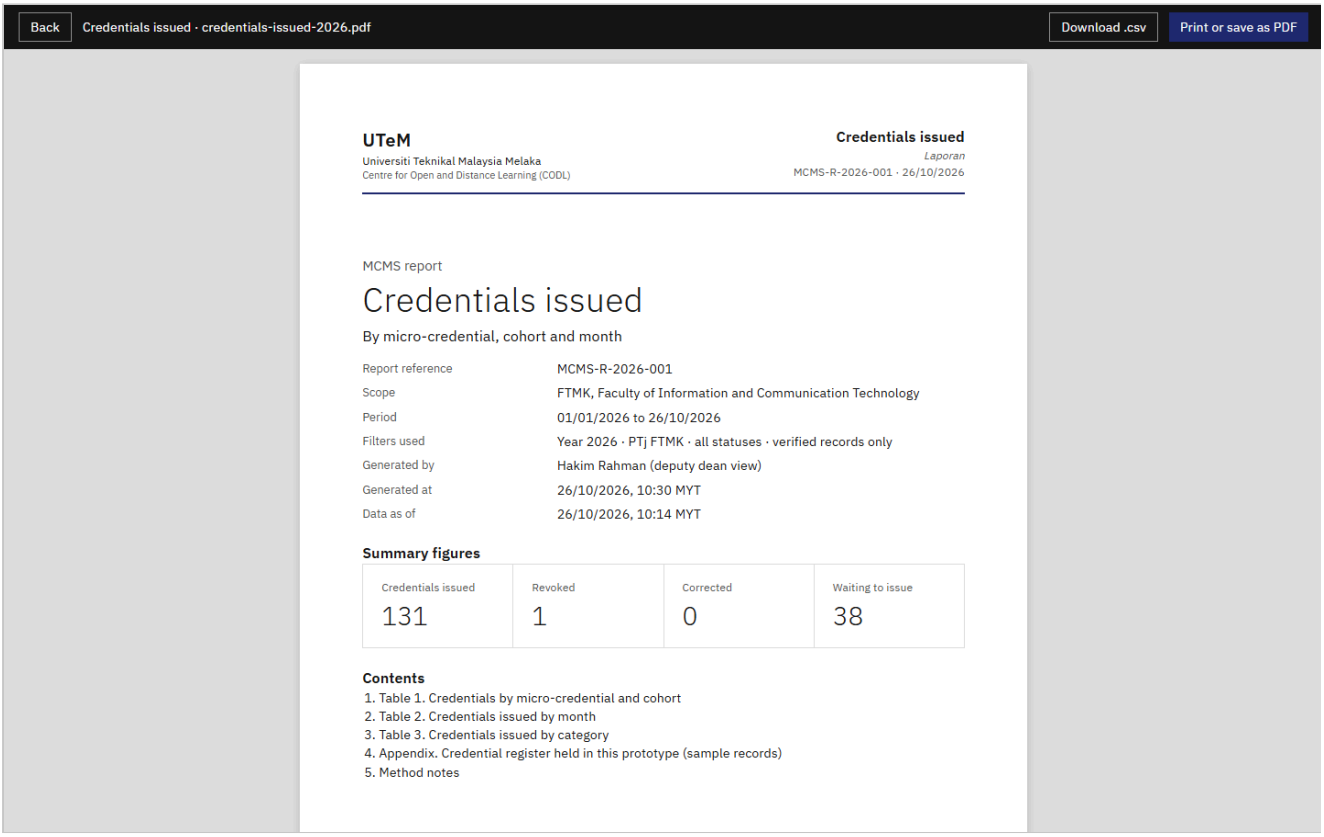


Figure 19: *Full report template (credentials issued), with PDF and CSV output.*

8.2 Content and Copy

Rule	Example
Plain, specific sentences with real figures	"Verifying issues module badges to 27 learners. Completion is confirmed separately."
Sentence case; no exclamation marks; no emoji	"Record Senate decision", not "RECORD SENATE DECISION!"
Never promise credit or transfer	"You may later apply to have it recognised; each transfer is decided case by case."
Name the category as in the Glossary	Unbundled Micro-Credential (UMC), Stand-alone Micro-Credential (SAMC), Micro-Credential Eksekutif (MCE)
No requirement codes in user text	Say the rule ("at least 1 completed cohort is required"), not its code
Say why a control is disabled	"Submit · 1 check failing", "Withdraw: the deadline 30/11/2026 has passed"
Status words exactly as in Section 4.8	"Provisionally redeemed", not "Used"
Both languages for every string	Missing BM or EN is marked on content fields and blocks template publishing

9. Traceability and Verification

9.1 Requirements to Screens

Table 18 maps each SRS functional module to the screens that deliver it. The per-requirement detail (actor, action, confirmation, result and blocking checks) is kept in `UIUX/v2/basis/srs-fr-inventory.md`.

Table 18: *SRS modules and the screens that deliver them*

SRS section	Module	Screens
3.1	Identity, roles and access	Sign-in, MFA, profile, CODL users and roles, settings, templates, reference data, adapter status
3.2	Category determination and proposal	Lecturer new proposal, proposal workspace, proposals and versions
3.3	Review, approval and versioning	Lecturer and expert review, deputy dean proposal review, Senate decision, decision pack, change to an approved version
3.4	Offering, catalogue and readiness	Lecturer create offering and readiness, deputy dean readiness, public catalogue and offering
3.5	Registration, enrolment and cohorts	Student register, registrations and payments, enrolment; lecturer cohorts
3.6	Learning platform connection	CODL UCreds exchange, import report, enrolment export
3.7	Assessment and achievement verification	Lecturer cohort results; deputy dean results and verify
3.8	Achievement Record	Student achievement record; deputy dean completion
3.9	Credential issuance and verification	Deputy dean completion and credentials; student credentials; public verify; certificate and credential documents
3.10	Quality Verification for SAMC	Lecturer QV evidence; deputy dean quality, QV record; QV package
3.11	ACR and ABC deposit	Student ledger and ACR consent; deputy dean ACR and deposit; ABC export
3.12–3.13	Credit transfer and single redemption	Student transfer; deputy dean transfer; transfer letter
3.14	Stackable degree pathway	Deputy dean pathways and plans; student plan; public pathways
3.15	Monitoring and CQI	Lecturer quality review; deputy dean quality review, periodic review, suspension
3.16	Records, audit trail and reporting	Reports, traceability, audit trail, retention, report documents
3.17	Notifications and integration hub	Notifications, preferences, adapter status, test mailbox

9.2 How the Design Was Verified

The prototype was checked in three ways before approval:

1. **Automated screen checks.** Every screen draws without script errors and every link, dialog, drawer and action it contains exists (the 114 screen entries (97 screens and 17 document entries) plus the prototype's two index pages, 51 dialogs and panels, 0 problems).
2. **Hand-off test.** A scripted walk through the real screens checks the role hand-offs of Section 7.2 (14 checks, all passing).
3. **Design review.** Three independent reviews compared every screen with the design basis and the SRS. They found 124 issues: 23 that broke a rule or blocked a flow, 70 missing states or confusing

behaviours, and 31 polish items. All rule and flow issues were fixed.

9.3 Corrections to the First Design

The first design (v1, Figma) was corrected against the SRS before the prototype was built. The main corrections, kept in the design rules, are:

First design	Corrected rule
Verifying results issued credentials	Verify results gives module achievements; confirming completion is a separate step that triggers issue (D-15)
Public pages claimed SAMC credit and QV beyond their validity	Credit and QV shown only while verified and valid for the cohort dates (BR-14, BR-23)
The deputy dean could revoke alone; "superseded" wording	Revoke needs the authorised body's decision reference; statuses Corrected, Replaced, Revoked
Review labels "Ready / Not ready"	Support, Amend, Do not support; conflict of interest declared first
Endorse allowed with open comments	Endorse blocked by open blocking comments
Dates and money in mixed formats	DD/MM/YYYY and RM 0.00
Public verification showed credit	Minimum fields: name, credential, issuer, date, status
Transfer could approve a line with no outcome match	Approve disabled for that line, with the reason (BR-30)
Phone-first student screens	Web first, reflowing to a phone
Neutral side navigation	Navy side navigation (owner's decision)

10. Assumptions, Open Items and Prototype Limits

10.1 Design Assumptions

ID	Assumption
UA-01	The "advisor" of a stackable plan (PRD US-48) is the deputy dean of the degree's faculty.
UA-02	A lecturer's scope is their own micro-credentials plus the reviews they are invited to.
UA-03	DESIGN.md (Carbon with uLearn Navy) is accepted as the UTeM corporate identity for MCMS (ir-identity; D-33, A-10).
UA-04	Participants sign in without MFA; every staff role that approves, verifies, issues, deposits or administers uses MFA.
UA-05	The withdrawal deadline is 20 days before the cohort ends until the fee and refund rules are set.
UA-06	FL-19 follows the demonstration scenario of PRD Section 7 (D-31). The design prototype and its v1.1 storyboard (Appendix D.21) still show the older flow that mixes three offerings; they are re-seeded to the PRD scenario when the prototype is next updated, and until then this specification controls.
UA-07	The step count of FL-19 is set when the prototype is re-seeded; the Release 1 prototype test follows the four acts of PRD Section 7.

10.2 Open Items Carried into Design

Item	Design treatment until it closes
OI-07 Retention periods	Retention categories shown with "TBD"; disposal requests wait for the Registrar.
OI-09 Transfer limits	Transfer case shows the limit check as "not configured yet; no limit is applied".
OI-11 ABC file format	ABC export marked "format not final"; columns shown can change.
OI-12 Fees and refunds	Receipts and withdrawals say refund rules are not decided yet.
OI-13 Completion window	Setting shown as TBD (global default, an offering may set its own); no window reminder is sent while no period is set.

10.3 Prototype Limits

- The prototype runs in a browser with sample data kept for the browser tab; "Reset data" restores the starting state. It has no server, accounts or real integrations.
- Screens show English text; the language switch works and is recorded, but BM strings are not written yet.
- The QR code on certificates is a drawn placeholder; signatures and hashes are test values.
- Excel export is marked Release 2; CSV and PDF are available.
- Screens were checked automatically at desktop width; the phone layout follows the responsive rules but has not been reviewed screen by screen.
- One sample learner appears in two Cloud Native Development cohorts; it does not affect any flow.
- The prototype flow FL-19 and its storyboard in version 1.1 predate the rewrite of Section 7.1 (UA-06).

Appendix A. Using the Prototype

The prototype is the living part of this specification. It runs from the project folder without installation:

Task	How
Start it	Run `py UIUX/v2/tools/serve.py 8731` and open http://localhost:8731
Choose a person	The Personas page opens first: student, lecturer, deputy dean, CODL, industry expert, public visitor, or the real sign-in screens
Walk a process	Flows (bottom left) lists the 26 flows; Start shows a step bar with what to click next
See what is clickable	Hotspots, or the H key, outlines every clickable element
Start again	Reset data restores the sample records
Open a document	Any download or print button, or the document index at `#/doc/index`
Check it	`node UIUX/v2/tools/check.js` (screens and links), `clickcheck.js` (dialogs), `scenario.js` (hand-offs), `pdfs.js` (sample PDFs)

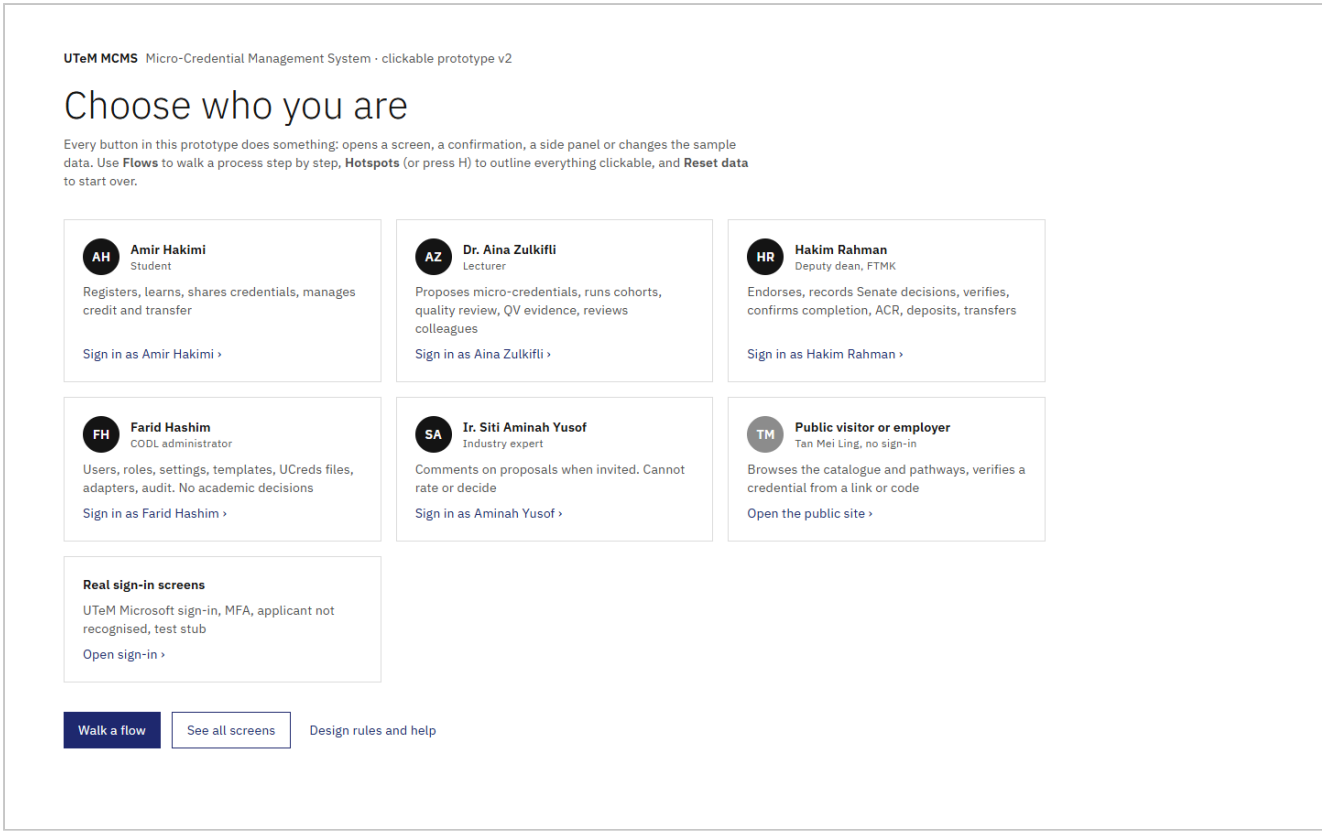


Figure 20: *Prototype start page: choose a persona.*

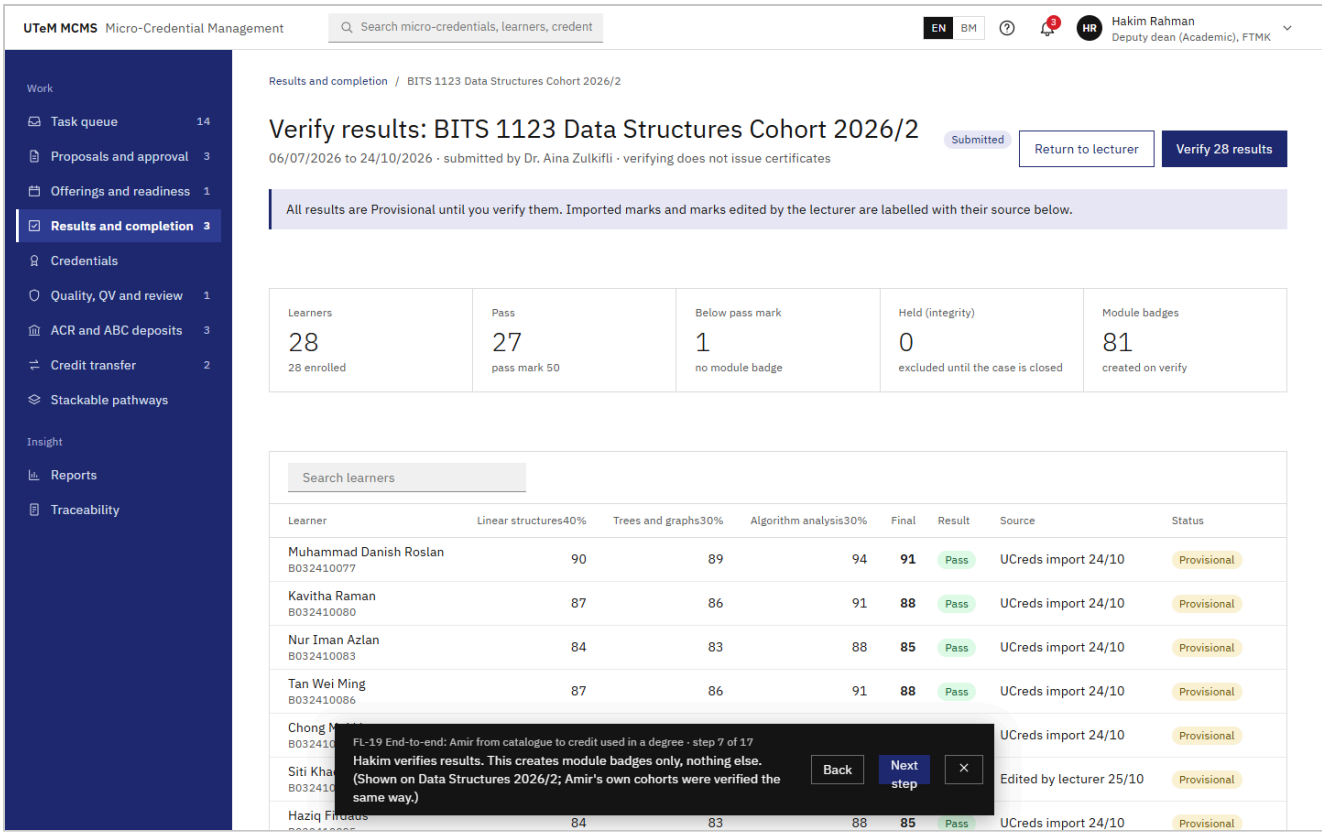


Figure 21: *A flow in progress: the step bar explains the step and offers Back, Next step and Exit.*

The pictures of Appendices B to D are captured from the prototype by `UIUX/v2/tools/gallery.js` and laid out by `UIUX/v2/tools/specgallery.js`, so they can be regenerated whenever the prototype changes.

Appendices B to D. Captures of the Design Prototype

Appendix B (every screen with its tabs, wizard steps, pop-ups, warnings, side panels and menus), Appendix C (every page of every printable document and report template) and Appendix D (a storyboard of every step of every flow) are screen captures of the design prototype. They are not repeated in this version, which changes the specification text only. They remain valid in version 1.1 (MCMS_UIUX_V1.1_Oct_2026), with one exception: Appendix D.21, the storyboard of FL-19, is superseded by Section 7.1 of this version (see UA-06 in Section 10).